# Stated or Sampled? Post-Training Packs a Language Model's Differential Diagnosis into a Single Call

Jairam Hathwar
University of Washington

Draft, October 26, 2026

**Abstract**

A language model's differential diagnosis can be read from the list it states in one answer or from the spread of its answers over repeated samples, the readout behind pass@$k$, self-consistency and sampling-based uncertainty. We measure how post-training changes both readouts on DDXPlus, whose synthetic patients carry a rule-based reference differential, with base/post-trained pairs from three open families at 7–32B (Qwen2.5 and Qwen3, every released stage of OLMo 3 7B and OLMo 3.1 32B, and Mistral-7B), zero- and few-shot prompts, and real rare-disease cases. Post-training largely removes the variation between calls. Sampled answers collapse (1.3–2.4 distinct top answers per 32 samples at $T$=0.7, against 3.8–4.9), and once every model sees worked examples the post-trained models lose 14–30 points of pass@32; for one of two models tested, sampling at a higher temperature recovers most of this. One stated list moves closer to the reference, but only by naming conditions with more reference mass, while its probabilities among those conditions move away, and only by about as much as averaging five base-model lists; the gain is robust for Qwen, weak for OLMo 3.1 32B and Mistral, and absent for OLMo 3 7B. Post-training also shifts which severe conditions are named, in both directions, rather than dropping them overall. Majority voting stops helping and agreement between samples weakens as an error signal; averaging the ranks of agreement and stated probability, with no labels, recovers most of what a labeled combination gains. Qwen3-8B's thinking mode reverses much of this: it beats its base model at pass@32 and restores agreement as the better error signal, though its answers stay narrow. Studies of diagnostic breadth should report which readout they use and whether the model reasons first.

## 1 Introduction

Clinicians reason over a differential diagnosis, a ranked set of conditions consistent with the presentation, and the conditions they keep in view determine which tests are ordered and which dangerous possibilities are ruled out. Language-model evaluations in medicine nonetheless score a single answer: multiple-choice accuracy, top-1 diagnosis, or at most pass@$k$ for a small $k$. Post-training, whether general instruction tuning and RLHF or domain-specific medical reinforcement learning and distillation, is reported to improve these scores [2, 10, 19].

A model's differential can be read in two ways: from the ranked list it states in one answer, or from the distribution of its answers over repeated samples. Much current practice uses the second. Pass@$k$, self-consistency voting [35] and sampling-based uncertainty such as semantic entropy [9] all draw on repeated samples, and so does the evidence that reinforcement learning sharpens models, which win at pass@1 while their base models solve more problems given many samples [26, 37, 39]. The two readouts agree for a model that samples from the distribution it states. Work outside medicine suggests that aligned models do not. They can verbalize distributions they fail to sample

from [24, 40], their verbalized confidence is better calibrated than their token probabilities [34], and their samples can homogenize enough to defeat sampling-based uncertainty [20]. Jang et al. [11] show this directly for survey-response distributions: instruction-tuned models sample worse than their own base models, more so at each post-training stage, while still describing the target accurately in a single call. We measure what this means for differential diagnosis, where a reference differential and a true diagnosis are available for every case. This is a replication and extension of that work in a clinical setting, not a new phenomenon; one aligned answer can also cover what many base-model samples spread out [16].

The design that holds the pretrained model fixed and varies only the post-training: (i) two base/instruct pairs, Qwen2.5-7B and Qwen3-8B, which isolate general post-training; (ii) three public medical models fine-tuned from the same parent, Qwen2.5-7B-Instruct: AlphaMed (rule-based RL), HuatuoGPT-o1 (SFT then PPO), and m1 (SFT distillation), plus AlphaMed's companion checkpoint trained from the base model; (iii) every released stage of OLMo 3 7B [33], from one base model through SFT, DPO and RL, plus a checkpoint trained with RL directly from the base; (iv) at 32B, Qwen2.5-32B with its Instruct model and every released stage of OLMo 3.1 32B. We draw 32 samples per case. On DDXPlus [8], which ships a reference differential for every synthetic patient, we score both readouts against that reference; on rare-disease cases from Phenopacket Store [6] we score accuracy and stated confidence.

**Findings.**

1. **Post-training collapses the sampled differential.** In both Qwen families at 7–8B the post-trained model is far better on its first answer (+13 to +27 points) but its pass@$k$ curve crosses the base model's at $k$≈4–8, and at $k$=32 the base model leads by 10–15 points; at 32B the lead is smaller (4.0–4.9 points), though it grows to 15–19 points when every model sees three worked examples (§4.10). A third family, Mistral-7B prompted with worked examples, shows the same crossing, also with example lists from another model instead of the reference, whereas Qwen3-8B with its thinking mode on beats its base model at $k$=32 (§4.12, §4.13, §4.11). Part of the narrowing is sharper sampling: Qwen2.5-7B-Instruct sampled at $T$=1.3 matches its base model's pass@32 at $T$=0.7, while Qwen3-8B stays below it even at $T$=1.6. Post-trained models give 1.3–2.6 distinct top answers per 32 samples, against 3.8–8.3, and can't-miss conditions drop out of their samples (§4.1, §4.2, §4.5). Which stage narrows the samples most depends on the recipe: the RL stage applied after SFT and DPO in OLMo 3 7B (19 points of pass@32), SFT in OLMo 3.1 32B (12 points), where the RL stage costs only 3 (§4.4).

2. **The stated differential improves, because one call now does the averaging.** A single stated list from the post-trained Qwen models at 7–32B is closer to the reference differential than the base model's (total-variation distance −0.04 to −0.09), covers 6–11 points more of its mass, and names severe conditions 9–14 points more often on average, while their samples move away from the reference. The same holds for OLMo 3.1 32B. The gain lies in the choice of conditions (the probability misallocated among named conditions rises), and the mean of five base-model lists is about as close to the reference as one post-trained list (§4.14). Severe conditions are reshuffled rather than dropped: after FDR correction 2–7 of the 17 are named significantly less often and 0–6 more often, never more declines than chance (§4.9). The gain survives restricting to complete lists, removing the true pathology, stratifying by top-1 correctness and correcting for each readout's floor (§4.9). OLMo 3 7B's five-item lists do not improve, but its stated distribution does when asked for a probability on every condition (§4.6, §4.8).

3. **Sampling-based methods lose their value; which signal to trust is model-dependent but cheap to check.** Majority voting over 32 samples gains base models 3–8 points at $T$=0.7 and the general post-trained models at most 1.9; models that reason before answering still gain 4–5. For post-trained Qwen models, the top three of one stated list beats the three most frequent of 32 sampled answers by 6–14 points. Agreement between samples becomes a weaker error signal in every family; the stated probability overtakes it for Qwen3-8B and OLMo-3-7B-Instruct but not for the other three general post-trained models. Averaging the two signals' ranks, with no labels, beats either signal for post-trained models (held-out AUROC 0.746 against 0.695 for agreement and 0.671 for the stated probability); a logistic model fitted on 100 labeled cases adds little (0.753). Qwen3-8B's thinking mode returns to the base-model pattern (§4.13). Neither readout tracks how ambiguous a case truly is (§4.7, §4.15).

4. **On real rare-disease cases the shifts replicate, though the top-three comparison does not flip.** Voting gains fall from 3.1–4.6 points for base models to 0.4–1.5 after post-training, and agreement weakens as an error signal until it ties the stated probability; 32 samples still beat one list (§4.18).

5. **Medical post-training adds accuracy, not breadth, and raises overconfidence on rare disease.** AlphaMed improves on its parent at every $k$ without widening its sampled differential; its companion checkpoint trained from the base model keeps most of the base model's diversity (§4.16). With accuracy near 2%, medical RL raises stated confidence from 0.51 to 0.67–0.71, and every OLMo stage states 0.73–0.84 at about 1% accuracy (§4.17).

6. **RL from the base model need not collapse the samples.** GRPO from Qwen3-8B-Base nearly matches the official post-trained model's top-1 accuracy while keeping more than twice its sampled diversity, and a group-level reward narrows the samples less than standard RLVR at equal accuracy (§5).

## 2 Related work

**What RL does to a model's output distribution.** Yue et al. [39] find that RL with verifiable rewards improves pass@1 on math and code while base models win at large $k$. Wu et al. [37] argue that RLVR mostly reweights solutions already in the base model's support. Oh et al. [26] document a "sharpening tax" across 14 base/post-trained pairs on agentic benchmarks. In contrast, Cheng et al. [4] report that RL expands the boundary on domains under-represented in pretraining. In medicine, the only pass@$k$ analysis we know of studies vision-language models and finds that RL mainly sharpens [12]. All of these measure the sampled distribution; we also measure the stated one.

**Diversity loss from alignment.** Instruction tuning and RLHF reduce output diversity [15, 25, 36], a loss that grows with scale for SFT [32]. Related work explains this as typicality bias in preference data [40] and documents cross-model homogeneity [13]. Zhang et al. [40] also track diversity across the SFT, DPO and RLVR checkpoints of Tülu 3 on creative writing; we do the same for OLMo 3 against a clinical reference distribution.

**Stated versus sampled distributions.** Meister et al. [24] find that language models asked to verbalize a group's opinion distribution match it better than their samples or log-probabilities

do, and that they "struggle to sample from their own distribution"; they study instruct and API models on opinion surveys, which have no correct answer. Verbalized sampling [40] prompts aligned models to state a distribution over candidate responses and recovers part of the diversity lost in post-training. Tian et al. [34] show that RLHF models' verbalized confidence is better calibrated than their token probabilities. Liu [20] report that aligned models' samples often fall into a single semantic cluster, leaving sampling-based uncertainty [9] without signal on those questions; in a Zephyr base/SFT/DPO ablation the collapse appears at DPO. Closest to our question, Jang et al. [11] show on synthetic and survey targets that instruction-tuned models sample worse than their own base models in every family they compare, that the gap grows at each OLMo-2 post-training stage, and that the same models describe the target accurately in one call. Verbalized-confidence studies reach mixed conclusions: sample consistency usually discriminates better than verbalized confidence [38], including for medical diagnosis [30]. Our stated readout is the verbalized top-$k$ elicitation of Tian et al. [34], and hybrids of consistency and verbalized confidence usually beat either [21, 28, 38]. We add a clinical task with a reference differential and a true diagnosis, two pipelines at 7–32B, and the consequences for aggregation and uncertainty.

**Medical reasoning models and calibration.** Medical post-training recipes include verifier-guided SFT and PPO [2], minimalist rule-based RL [19], and test-time-scaling distillation [10]. Reward models and RL tend to favor confident answers [5, 17], and reasoning models are often overconfident [23]. LLMs have been evaluated as differential-diagnosis aids on clinicopathological cases [14, 22]. We show that this overconfidence appears in public medical descendants of one parent, on rare disease.

**Differential diagnosis benchmarks and distributional RL.** DDXPlus [8] was introduced for automatic diagnosis systems trained on differentials [7]. Recent LLM evaluations on it score hierarchical top-$k$ accuracy [18] or interactive diagnosis [29]; the dataset's own metrics compare predicted and reference differentials as sets. Puri et al. [27] train models with GRPO to output several answers per forward pass, rewarding coverage of a gold *set* and set-level calibration, and evaluate on DDXPlus. We instead compare stated and sampled differentials against the reference *distribution* across public post-trained model families. Our group-level reward operates on the distribution of independently sampled answers (§5).

## 3 Experimental setup

**Tasks.** **DDXPlus** [8]: synthetic patients produced by a rule-based medical knowledge base. Each patient has a true pathology and a reference differential, a probability distribution over the 49 conditions. We use 980 test patients, 20 per condition. Each patient's coded symptoms and history are rendered as a short vignette; we fix known mistranslations in the English evidence descriptions. The prompt lists all 49 conditions and asks for the five most likely, with probabilities, as JSON. **Phenopacket Store** [6] (v0.1.27): published rare-disease cases annotated with HPO phenotypes and an OMIM diagnosis. We use a disease-balanced set of 1,401 cases covering 749 diseases (at most two per disease). Each case is a vignette of present and absent phenotypes, and the model is asked for an open-ended differential. Answers are graded by mapping names to Mondo; we credit an exact match, and separately a "related" match to a specific ancestor group or subtype.

**Models.** Table 1. All checkpoints are public. For the medical models we verified the parent model from each released configuration.

| Model | Parent | Post-training | Role |
|---|---|---|---|
| Qwen2.5-7B | n/a | none (pretrained) | base |
| Qwen2.5-7B-Instruct | Qwen2.5-7B | general SFT + RLHF | instruct; medical parent |
| AlphaMed-7B [19] | Qwen2.5-7B-Instruct | rule-based RL | medical |
| HuatuoGPT-o1-7B [2] | Qwen2.5-7B-Instruct | SFT + PPO (verifier) | medical |
| m1-7B-23K [10] | Qwen2.5-7B-Instruct | SFT distillation | medical |
| Qwen3-8B-Base | n/a | none (pretrained) | base |
| Qwen3-8B | Qwen3-8B-Base | general post-training | instruct |
| Qwen2.5-32B | n/a | none (pretrained) | base |
| Qwen2.5-32B-Instruct | Qwen2.5-32B | general SFT + RLHF | instruct |
| OLMo-3-1025-7B [33] | n/a | none (pretrained) | base |
| OLMo-3-7B-Instruct-SFT | OLMo-3-1025-7B | SFT | stage 1 |
| OLMo-3-7B-Instruct-DPO | OLMo-3 SFT | DPO | stage 2 |
| OLMo-3-7B-Instruct | OLMo-3 DPO | RLVR | stage 3 (instruct) |
| OLMo-3-7B-RL-Zero-General | OLMo-3-1025-7B | RL (LM judge) | RL from base |
| OLMo-3-1125-32B | n/a | none (pretrained) | base |
| OLMo-3.1-32B-Instruct-SFT | OLMo-3-1125-32B | SFT | stage 1 |
| OLMo-3.1-32B-Instruct-DPO | OLMo-3.1 SFT | DPO | stage 2 |
| OLMo-3.1-32B-Instruct | OLMo-3.1 DPO | RLVR | stage 3 (instruct) |

Table 1: Models studied. Each pair or family shares its pretrained weights, so differences reflect post-training. OLMo 3 and OLMo 3.1 release every stage of their instruct pipelines.

**Sampling.** We draw $n$=32 samples per case with vLLM, at $T$=0.7, top-$p$ 0.95, and at $T$=1.0, top-$p$ 1.0 (sampling from the model's own distribution). All models use their chat template and the same prompt; OLMo-3-1025-7B ships without one, so it gets its SFT checkpoint's template. Completions are capped at 4,096 tokens (8,192 for m1, whose long reasoning otherwise truncates), and generation stops at the end-of-turn token. The 32B models are sampled at $T$=0.7 only; the OLMo 3.1 32B SFT and DPO checkpoints, stored in float32, are run in bfloat16 like the other stages. We take the last valid JSON list in each completion. Unparsable completions count as incorrect; this affects mainly base models, so base-model pass@$k$ is a lower bound at large $k$, while post-trained gains at small $k$ are partly format. §4.3 reports format-controlled comparisons.

**Metrics.** *pass@k*: the unbiased estimator of Chen et al. [3] on the top-1 answer of each sample. *Distinct* and *entropy*: the number of distinct top-1 diagnoses across a case's 32 samples, and the entropy of their empirical distribution, averaged over cases. On DDXPlus, $TV$: total-variation distance between this empirical distribution and the reference differential. *Coverage@k*: the expected share of the reference differential's probability mass covered by the distinct top-1 answers in $k$ random samples. *Calibration*: expected calibration error (10 bins) of the stated top-1 probability against top-1 correctness. These metrics describe the *sampled* readout. §4.6 defines the corresponding metrics for the *stated* readout, the list in a single completion. Differences come with 95% paired-bootstrap intervals over cases (2,000 resamples) and are computed from unrounded values. We report many intervals without multiplicity correction; we treat a non-significant difference as "no detectable change", with the interval bounding its size, not as equivalence.

| Pair | Model | pass@1 | pass@32 | Distinct | Entropy | TV↓ | Cov@32 | ECE↓ |
|---|---|---|---|---|---|---|---|---|
| Qwen3-8B, $T$=1 | base | 0.282 | **0.789** | **6.60** | **1.31** | **0.712** | **0.407** | 0.214 |
| | post-trained | **0.556** | 0.643 | 1.53 | 0.18 | 0.783 | 0.236 | **0.061** |
| Qwen2.5-7B, $T$=1 | base | 0.239 | **0.766** | **8.23** | **1.58** | **0.715** | **0.422** | 0.303 |
| | instruct | **0.464** | 0.663 | 2.65 | 0.49 | 0.761 | 0.290 | **0.137** |
| Qwen3-8B, $T$=0.7 | base | 0.388 | **0.730** | **3.76** | **0.78** | **0.750** | **0.322** | 0.121 |
| | post-trained | **0.558** | 0.607 | 1.27 | 0.11 | 0.793 | 0.217 | **0.063** |

Table 2: DDXPlus, 32 samples per case (980 cases). The reference differential has mean entropy 1.87. Distinct and entropy refer to the top-1 answers across samples; Cov@32 is the share of reference probability mass covered by those answers. ECE is for the stated top-1 probability.

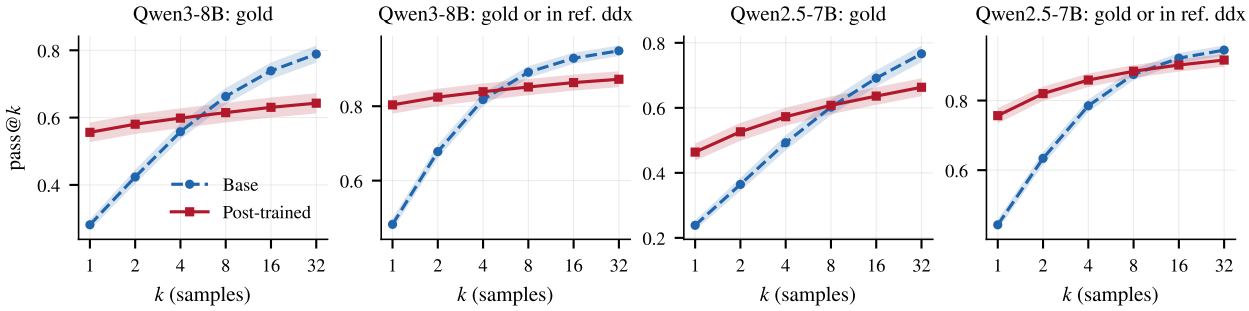



Figure 1: pass@$k$ on DDXPlus at $T$=1 (980 cases, 32 samples), base model (dashed) vs. its post-trained version (solid), with 95% case-bootstrap bands. "Gold": the top-1 answer is the true pathology; "gold or in ref. ddx": it is the true pathology or appears in the reference differential. In both families the post-trained model wins at small $k$ and loses at large $k$.

# 4 Results

## 4.1 General post-training trades coverage for top-1 accuracy

Table 2 and Figure 1 compare each base model with its post-trained counterpart on DDXPlus. In both families the post-trained model is much better on its first answer: +27.5 points for Qwen3 and +22.5 for Qwen2.5 at $T$=1 (intervals $[+25.3, +29.6]$ and $[+20.7, +24.4]$). The advantage shrinks with $k$ and reverses: the curves cross at $k$≈4–8, and at $k$=32 the base model leads by 14.6 points (Qwen3, $[+12.2, +17.1]$) and 10.3 points (Qwen2.5, $[+7.8, +12.8]$). The pattern is the same at $T$=0.7 for Qwen3, crossing at $k$=4 with a base lead of 12.2 points at $k$=32 ($[+9.9, +14.9]$). Unparsable base-model samples count as wrong, so these gaps are lower bounds. Qwen2.5 at $T$=0.7 behaves the same way: +12.6 points at $k$=1 $[+10.7, +14.5]$ and a base lead of 9.7 at $k$=32 $[+7.3, +12.0]$.

## 4.2 The sampled differential collapses, while top-1 calibration improves

The gap at large $k$ reflects a collapsed *sampled* differential. Across 32 samples, the post-trained Qwen models name 1.3–2.6 distinct diagnoses per case, against 3.8–8.2 for their base models. Their answer entropy is 0.11–0.49, against 0.78–1.58 for base models and 1.87 for the reference differential (Table 2). Sampling at $T$=1 widens the base models' answers but barely changes the post-trained models' (Qwen3 entropy 0.11 at $T$=0.7, 0.18 at $T$=1), so the collapse lives in the model's own distribution rather than in the decoding settings. As a result, post-trained models cover less of the

reference differential: 0.24 vs. 0.41 of its probability mass at $k$=32 for Qwen3, and 0.29 vs. 0.42 for Qwen2.5. Their top-1 calibration, however, is markedly better (ECE 0.06 vs. 0.21 and 0.14 vs. 0.30). On every single-answer criterion (accuracy, small-$k$ pass@$k$, calibration), the post-trained model therefore looks strictly better, and the loss of breadth in its samples is invisible. Whether the breadth is lost from the model's stated differential as well is a separate question (§4.6).

**Which diagnoses drop out?** For each case and each condition that the reference differential weights at least 0.1, we ask whether the model names that condition as its top answer in at least one of its 32 samples. We call this *considering* the condition. We group the 49 conditions by DDXPlus severity (1 = most severe). Post-training lowers consideration in every tier (Table 3), and the severe tier (DDXPlus severity 1–2, the dataset's label rather than a clinical consensus list) is not spared. This metric needs a chance reference, though: a guesser that picks uniformly among the 49 conditions scores about 0.48 on it, and one that draws from the average reference differential 0.56–0.61, above every model, base models included. Considered@32 therefore measures the breadth of the sampled answers, not clinical judgement; we assess can't-miss conditions on the stated list instead (§4.9). For the 17 severity-1–2 conditions (e.g. NSTEMI/STEMI, acute pulmonary edema, anaphylaxis, pulmonary embolism, unstable angina), consideration falls from 0.45 to 0.27 for Qwen3 and from 0.51 to 0.34 for Qwen2.5 at $T$=1. The differences are $-18.4$ $[-20.8, -15.9]$ and $-17.0$ $[-19.7, -14.2]$ points. Some diagnoses disappear from the top answer almost entirely. Qwen2.5-7B-Instruct names unstable angina as its top answer in one of 107 eligible cases, where its base model does in 47%. When unstable angina is the true pathology, pass@32 falls from 1.00 to 0.00. Stable angina (0.89 to 0.06) and myocarditis (0.41 to 0.00) follow the same pattern. These conditions are not forgotten: they still appear lower in the listed differential (unstable angina in 50% of eligible cases, against 95% for the base model). The model ranks them but no longer commits to them. At the same time, post-trained models concentrate on a few salient "attractor" diagnoses. Qwen2.5-7B-Instruct gives NSTEMI/STEMI as its top answer in 13.1% of samples, against a reference rate of 4.4% and 6.4% for its base model. It also over-predicts pulmonary embolism (8.2% vs. 3.2%) and pneumonia (8.0% vs. 2.4%). Chest-pain presentations thus collapse onto the most dangerous candidate. That raises its accuracy (NSTEMI pass@1 0.99 vs. 0.44) at the cost of the alternatives, a bias toward the most dangerous candidate rather than a differential.

### 4.3 Robustness to prompt and output format

**Prompt.** We repeat the Qwen3-8B pair with two other prompts (Table 4). *Plain* drops the system prompt and rewords the instructions. *CoT* asks the model to reason step by step before answering; Qwen3's thinking mode stays off in all runs. The plain prompt reproduces the pattern: the post-trained model gains $+23.0$ $[+20.9, +25.1]$ points at $k$=1 and loses $-7.1$ $[-9.3, -5.0]$ at $k$=32 ($T$=0.7; at $T$=1, $+34.2$ and $-8.7$ $[-11.0, -6.4]$), with 1.3–1.5 distinct answers against 3.7–5.6. Asking for reasoning changes the picture. With CoT, the post-trained model's pass@32 rises from 0.607 to 0.835: it leads the base model at every $k$ at $T$=0.7 ($+4.7$ $[+2.9, +6.6]$ at $k$=32) and ties it at $T$=1 (0.0 $[-2.1, +1.9]$). Its samples remain far less diverse than the base model's (2.1–2.2 vs. 3.9–6.3 distinct answers), so reasoning recovers coverage at $k$=32 without restoring sampling diversity. The coverage loss is therefore largest when the model answers directly, as in most clinical decision-support deployments, and reasoning at inference time mitigates it. This matches the medical models: the reasoning-trained descendants widen the differential (§4.16).

**Output format.** Base models more often give invalid answers: unparsable, or with a top answer outside the 49 conditions. This affects 12% of Qwen3-8B-Base samples at $T$=0.7 and 25% at $T$=1

| Comparison | Considered@32 (base → post-trained) Severe (1–2) | Moderate (3) | Mild (4–5) |
|---|---|---|---|
| Qwen3-8B, $T$=1 | 0.45 → 0.27 | 0.50 → 0.32 | 0.45 → 0.20 |
| | −18.4 [−20.8, −15.9] | −18.1 [−20.7, −15.7] | −24.2 [−27.0, −21.4] |
| Qwen2.5-7B, $T$=1 | 0.51 → 0.34 | 0.50 → 0.37 | 0.38 → 0.23 |
| | −17.0 [−19.7, −14.2] | −12.1 [−14.7, −9.7] | −15.0 [−17.4, −12.6] |
| Qwen3-8B, $T$=0.7 | 0.34 → 0.24 | 0.44 → 0.29 | 0.31 → 0.18 |
| | −9.8 [−11.8, −7.9] | −15.8 [−18.2, −13.5] | −13.4 [−15.8, −11.1] |
| Qwen2.5-7B, $T$=0.7 | 0.40 → 0.28 | 0.41 → 0.32 | 0.28 → 0.17 |
| | −12.4 [−14.7, −10.2] | −9.1 [−11.2, −6.7] | −10.3 [−12.3, −8.3] |
| Instruct → AlphaMed, $T$=0.7 | 0.28 → 0.32 | 0.32 → 0.34 | 0.17 → 0.21 |
| | +4.3 [+2.0, +6.6] | +2.2 [+0.4, +4.0] | +3.9 [+2.4, +5.4] |
| Instruct → HuatuoGPT-o1, $T$=0.7 | 0.28 → 0.36 | 0.32 → 0.37 | 0.17 → 0.27 |
| | +8.4 [+6.1, +10.8] | +6.0 [+4.0, +7.9] | +9.5 [+7.5, +11.5] |
| Instruct → m1, $T$=0.7 | 0.28 → 0.41 | 0.32 → 0.41 | 0.17 → 0.29 |
| | +13.5 [+10.8, +16.2] | +9.7 [+7.2, +12.2] | +11.4 [+9.2, +13.6] |
| Base → AlphaMed (base-RL), $T$=0.7 | 0.40 → 0.39 | 0.41 → 0.40 | 0.28 → 0.27 |
| | −0.7 [−3.0, +1.7] | −0.4 [−2.6, +1.7] | −0.4 [−2.4, +1.7] |
| Base → `acc` GRPO (8B), $T$=0.7 | 0.34 → 0.31 | 0.44 → 0.42 | 0.31 → 0.27 |
| | −2.8 [−4.4, −1.3] | −2.1 [−3.7, −0.6] | −4.3 [−6.2, −2.5] |
| Base → `gm` GRPO (8B), $T$=0.7 | 0.34 → 0.32 | 0.44 → 0.43 | 0.31 → 0.28 |
| | −1.5 [−3.2, +0.1] | −1.3 [−3.0, +0.3] | −3.0 [−4.8, −1.2] |

Table 3: Share of (case, condition) pairs with reference probability ≥ 0.1 in which the condition is a top-1 answer in at least one of 32 samples, by DDXPlus severity. Second rows give differences in points with 95% case-bootstrap intervals. All rows use the 980 test cases; GRPO rows are seed 0. General post-training lowers consideration in every tier, including the can't-miss tier; medical post-training raises it; our GRPO from the base model loses at most 4 points, and with the group reward the severe-tier loss is not significant.

(Qwen2.5-7B: 1% and 21%), against at most 0.3% for the post-trained models. Invalid samples count as wrong, which deflates the base model at large $k$ but inflates the post-trained model's gain at small $k$. Among valid samples only, the post-trained models still gain +12.0 [+9.8, +14.1] (Qwen3, $T$=0.7), +18.7 [+16.6, +20.7] (Qwen3, $T$=1), +12.2 [+10.4, +14.0] and +16.8 [+15.0, +18.5] points (Qwen2.5) at $k$=1, so a quarter to a third of the raw gain at $T$=1 is format. On the cases where every sample of both models is valid, the large-$k$ loss persists: −15.2 [−21.9, −8.6] at $k$=32 for Qwen3 (105 cases) and −10.2 [−12.7, −7.6] for Qwen2.5 (748 cases), both at $T$=0.7. (At $T$=1 almost no case has 32 valid base samples.) The reported large-$k$ losses are thus lower bounds.

### 4.4 Which post-training stage narrows the differential?

The Qwen pairs compare a base model with the end of a post-training pipeline. OLMo 3 [33] releases every stage of its 7B instruct pipeline from one base model: supervised fine-tuning, then DPO, then RL with verifiable rewards on math, code, instruction-following and chat prompts (the released OLMo-3-7B-Instruct). It also releases RL-Zero checkpoints trained with RL directly from the base; we use the general one, trained on chat prompts with an LM-judge reward. (The RL-Zero-Mix checkpoint does not load in our vLLM version.) Table 5 gives the results.

The coverage loss replicates in OLMo. At $T$=0.7, every post-trained stage is worse than the base at $k$=32: by −14.6 [−17.5, −11.7] points after SFT, −9.2 [−12.1, −6.1] after DPO, −28.3

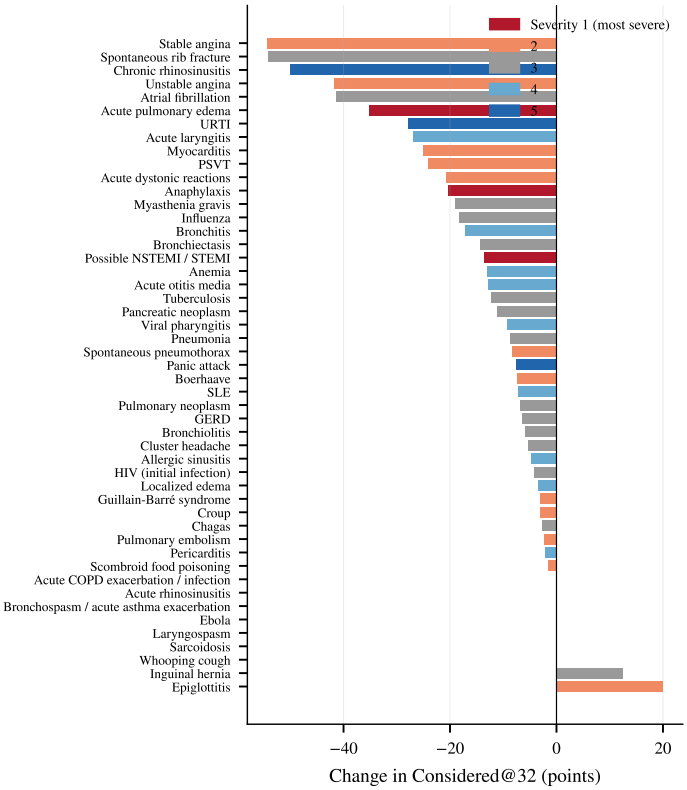



Figure 2: Change in Considered@32 per condition, Qwen2.5-7B-Instruct minus its base model ($T$=1, 980 cases), colored by DDXPlus severity (dark red = most severe). Conditions with fewer than five cases giving them reference probability $\geq 0.1$ are omitted. Angina, acute pulmonary edema, anaphylaxis and myocarditis are among the largest drops.

$[-31.5, -25.0]$ after RLVR, and $-13.3$ $[-16.1, -10.5]$ for RL-Zero. Diversity falls at every stage of the pipeline, from 4.93 distinct answers for the base to 4.01 after SFT, 3.30 after DPO and 2.37 after RLVR.

Compared stage to stage, DPO recovers part of what SFT lost at large $k$ ($+5.4$ $[+2.9, +8.0]$ points at $k$=32, and $+10.3$ $[+9.0, +11.6]$ at $k$=1) while narrowing the differential further ($-0.71$ distinct answers). The final RLVR stage is the largest single narrowing. Against the DPO model it loses $-19.1$ $[-21.8, -16.3]$ points at $k$=32, 0.92 distinct answers and 7.7 points of coverage@32, and it is also worse at $k$=1 ($-5.9$ $[-7.2, -4.6]$). It drops severe-tier Considered@32 to 0.25, a loss of $-16.1$ $[-18.6, -13.5]$ points relative to the base, against $-7.4$ for the DPO model. At $T$=1 the pattern is the same (RLVR vs. DPO: $-12.7$ $[-15.4, -9.9]$ at $k$=32).

RL applied directly to the base narrows the differential less. At $T$=1, RL-Zero shows no detectable loss of the base model's pass@32 ($-1.8$ $[-4.5, +0.8]$; clustered $[-5.8, +2.1]$) and 6.28 of its 7.53 distinct answers, and it loses 4.0 $[1.2, 6.9]$ points of severe-tier consideration, against 14.4 for the full instruct pipeline. At $T$=0.7 it still loses 13.3 points at $k$=32 (clustered $[7.3, 19.7]$), about as much as SFT.

The OLMo base model often breaks the output format: 40% of its samples are invalid at $T$=0.7 and 47% at $T$=1, while unparsable completions are 16–21% for the SFT model and at most 1.6% for the later stages. No case has 32 valid base samples, so the all-valid check of §4.3 is not possible here. Among valid samples, the post-trained models' gains at $k$=1 are small ($-1.9$ to $+4.0$ points

| | | pass@1 | | pass@32 | | Distinct | |
|---|---|---|---|---|---|---|---|
| Prompt | $T$ | Base | Post | Base | Post | Base | Post |
| default | 0.7 | 0.388 | **0.558** | **0.730** | 0.607 | 3.76 | 1.27 |
| default | 1.0 | 0.282 | **0.556** | **0.789** | 0.643 | 6.60 | 1.53 |
| plain | 0.7 | 0.352 | **0.583** | **0.727** | 0.655 | 3.72 | 1.28 |
| plain | 1.0 | 0.244 | **0.586** | **0.792** | 0.705 | 5.57 | 1.54 |
| CoT | 0.7 | 0.409 | **0.652** | 0.788 | **0.835** | 3.89 | 2.08 |
| CoT | 1.0 | 0.275 | **0.642** | 0.837 | 0.837 | 6.31 | 2.24 |

Table 4: Qwen3-8B-Base vs. Qwen3-8B on DDXPlus (980 cases, 32 samples) under three prompts. Bold marks the significantly better model (paired case bootstrap). Without reasoning, the base model wins at $k$=32; with CoT, the post-trained model matches or beats it, but its samples stay far less diverse.

| $T$ | Stage | pass@1 | pass@32 | Distinct | Entropy | Cov@32 | Severe C@32 |
|---|---|---|---|---|---|---|---|
| 0.7 | Base | 0.178 | **0.705** | **4.93** | **1.16** | **0.351** | **0.412** |
| | SFT | 0.222 | 0.559 | 4.01 | 0.87 | 0.257 | 0.373 |
| | SFT+DPO | **0.324** | 0.613 | 3.30 | 0.63 | 0.272 | 0.338 |
| | SFT+DPO+RLVR (Instruct) | 0.266 | 0.422 | 2.37 | 0.42 | 0.195 | 0.251 |
| | RL-Zero (RL from base) | 0.295 | 0.572 | 4.06 | 0.84 | 0.286 | 0.326 |
| 1.0 | Base | 0.128 | **0.713** | **7.53** | **1.66** | **0.398** | **0.469** |
| | SFT | 0.191 | 0.634 | 6.32 | 1.29 | 0.321 | 0.426 |
| | SFT+DPO | **0.313** | 0.688 | 4.48 | 0.87 | 0.312 | 0.410 |
| | SFT+DPO+RLVR (Instruct) | 0.272 | 0.561 | 3.58 | 0.68 | 0.259 | 0.325 |
| | RL-Zero (RL from base) | 0.279 | 0.695 | 6.28 | 1.21 | 0.367 | 0.429 |

Table 5: OLMo 3 7B post-training stages on DDXPlus (980 cases, 32 samples). Severe C@32 is severe-tier Considered@32 as in Table 3. Bold marks the best value in each block. Diversity falls at every stage of the instruct pipeline, and the RLVR stage loses the most coverage; RL from the base keeps the base model's pass@32 at $T$=1.

at $T$=0.7, +1.3 to +8.1 at $T$=1), so most of OLMo's raw gain at $k$=1 is format. Invalid samples count as misses and are excluded from distinct answers and entropy, so the base model's coverage and diversity are understated and the losses above are conservative.

Across these pipelines, then, the collapse of the sampled differential is not a necessary effect of RL. It accumulates over the instruction-tuning pipeline; in OLMo 3 7B its largest step is RL applied on top of SFT and DPO, but at 32B it is SFT (§4.5), so which stage narrows most depends on the recipe. RL applied directly to a base model (OLMo RL-Zero, AlphaMed's base-initialized checkpoint, and our GRPO runs in §5) narrows the differential less. On Phenopacket Store the instruct pipeline also loses coverage: every stage trails the base model at $k$=32 ($-2.0$ $[-3.0, -1.1]$ points after SFT, $-1.5$ after DPO, $-1.4$ after RLVR; all significant), while RL-Zero does not ($-0.6$ $[-1.4, +0.2]$), from a base model at 4.5% (§4.17).

### 4.5 Replication at 32B

We repeat the DDXPlus comparisons at $T$=0.7 with Qwen2.5-32B and its Instruct model, and with every released stage of OLMo 3.1 32B, whose base is OLMo-3-1125-32B; there is no 32B RL-Zero checkpoint (Table 6). The sampled differential collapses at 32B too, by about half as much. Post-training costs 4.9 [2.8, 7.1] points at pass@32 for Qwen2.5-32B and 4.0 [2.0, 5.9] for OLMo 3.1

| Model | pass@1 | pass@32 | Distinct | Cov@32 | Severe C@32 | TV one list↓ | TV samples↓ | Vote@32 gain |
|---|---|---|---|---|---|---|---|---|
| Qwen2.5-32B (base) | 0.500 | **0.795** | **3.78** | **0.352** | **0.35** | 0.726 | **0.726** | +7.0 |
| Qwen2.5-32B-Instruct | **0.604** | 0.746 | 1.68 | 0.276 | 0.29 | **0.674** | 0.759 | +1.9 |
| OLMo-3-1125-32B (base) | 0.326 | **0.796** | **4.32** | **0.347** | **0.39** | 0.746 | **0.742** | +8.4 |
| SFT | 0.461 | 0.674 | 2.94 | 0.265 | 0.27 | 0.736 | 0.776 | +3.6 |
| SFT+DPO | **0.584** | 0.784 | 2.29 | 0.293 | 0.32 | **0.694** | 0.755 | +2.1 |
| SFT+DPO+RLVR (Instruct) | 0.560 | 0.756 | 2.15 | 0.283 | 0.30 | 0.701 | 0.763 | +0.9 |

Table 6: 32B replication on DDXPlus ($T$=0.7, 980 cases, 32 samples). TV columns: distance to the reference differential of one stated list and of the sampled top-1 answers (§4.6). Vote@32 gain: accuracy of the majority answer over 32 samples minus single-call accuracy, in points (§4.7). Bold: best value per family.

32B (clustered $[-0.6, 10.2]$ and $[1.2, 6.8]$; neither survives correction), halves the number of distinct answers (3.78 → 1.68 and 4.32 → 2.15), and lowers severe-tier consideration by 6.1 and 8.4 points. For Qwen the loss holds on the 421 cases where every base sample is valid ($-4.8$ $[-7.8, -1.7]$); the OLMo 32B base gives invalid answers on 27% of samples. With few-shot prompts, which nearly remove invalid answers, both losses are three to five times larger (§4.10). The stage attribution differs from 7B. In OLMo 3.1 32B the largest narrowing is SFT (pass@32 $-12.1$ $[-14.4, -9.9]$, clustered $[-18.0, -6.8]$, distinct answers $-1.39$, severe tier $-11.9$); DPO recovers most of it (+10.9 against SFT), and the final RL step is small ($-2.8$ $[-4.3, -1.1]$ at $k$=32, clustered $[-5.3, -0.5]$, $-0.14$ distinct answers). OLMo 3.1 32B and OLMo 3 7B were post-trained with different recipes, so the stage that narrows the samples most is a property of the pipeline, not of SFT, DPO or RL as such. Under this prompt the stated readout improves in both 32B families (§4.6; with few-shot prompts only Qwen2.5-32B's does robustly, §4.10), and voting stops paying off: majority voting over 32 samples gains the base models 7.0 and 8.4 points and the Instruct models 1.9 and 0.9 (the latter n.s.).

### 4.6 The stated differential does not collapse

Every completion also *states* a differential: up to five conditions with probabilities. We read it directly. For each valid sample we renormalize its stated probabilities over the conditions it names (uniform if the probabilities are missing or invalid) and compare this single list with the reference differential: its total-variation distance to the reference, the reference mass of the conditions it names, and, for each severe condition with reference probability at least 0.1, whether the list names it. For the sampled readout we use the empirical distribution of top-1 answers over the valid samples and, for the same budget of five names, the reference mass covered by the distinct top answers of five random samples. Two oracles bound what each format can achieve: the reference's own top five stated as a list, and as many draws from the reference as the model has valid samples (Table 7, Figure 3).

For the Qwen models the two readouts move in opposite directions under post-training. In both families and at both temperatures, the post-trained model's single list is closer to the reference than the base model's (TV $-0.043$ to $-0.086$, all intervals exclude zero), covers more of its mass (+6.1 to +10.8 points), and names severe conditions more often (+9.1 to +13.4 points). Its sampled top answers move farther from the reference (+0.029 to +0.071), and five of them cover less ($-1.9$ to $-6.0$ points). Within each post-trained model the list beats its own samples by 0.04–0.09 in TV; within each base model the gap is about zero or favors the samples. The CoT prompt keeps the list's advantage (TV $-0.042$, severe naming +8.1 points) and removes the loss in the samples (+0.002, n.s.). This quantifies the observation of §4.2 that conditions dropping out of the top

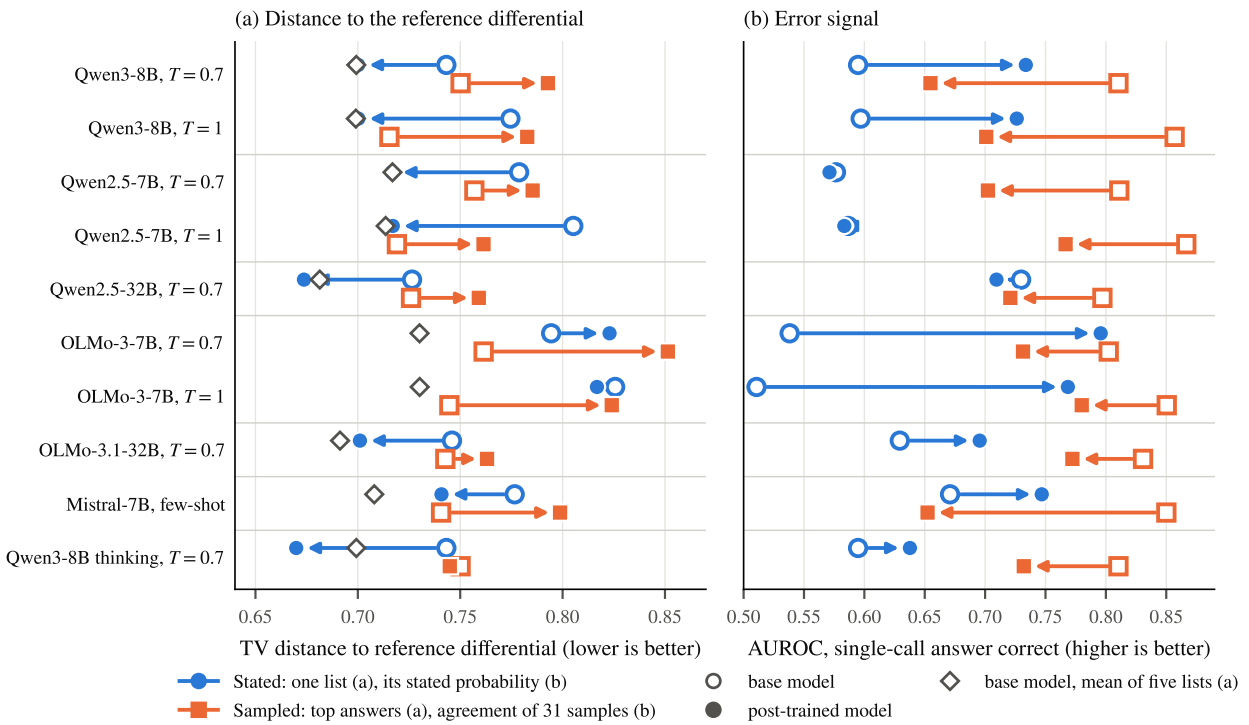



Figure 3: Base (hollow) → post-trained (filled) on DDXPlus (980 cases; OLMo: base → final Instruct; 32B models at $T$=0.7 only). (a) TV distance to the reference differential of one stated list (blue circles) and of the sampled top-1 answers (orange squares). (b) AUROC for the single-call answer being correct, of its stated probability (blue) and of agreement across 31 further samples (orange).

answer remain lower in the list.

At 32B the pattern holds in both families (Table 6): the post-trained lists are closer to the reference (TV −0.053 for Qwen2.5-32B-Instruct, −0.052 and −0.045 for the OLMo 3.1 DPO and Instruct models) and cover 5.6–6.7 points more of it, while the samples move away (+0.014 to +0.034). Within every post-trained 32B model the list beats its own samples by 0.04–0.09 in TV, against about zero for both base models. OLMo 3 7B is the exception with the five-item prompt. Its DPO and Instruct models' lists are no closer to the reference than the base model's (TV +0.000 and +0.028 at $T$=0.7; −0.026 and −0.005 at $T$=1) and cover at most 5 points more of it; only the naming of severe conditions improves (+6.1 to +12.8 points). Asked for a probability on every condition, its Instruct model's stated distribution is closer to the reference than the base model's (−0.032 [−0.038, −0.027]; §4.8), so the exception is specific to the short list. RL-Zero's lists improve slightly (TV −0.020 and −0.039). The medical descendants change little relative to their parent (TV −0.012 for AlphaMed, −0.014 for HuatuoGPT-o1, +0.017 for m1). In absolute terms every model is far from the reference: a single list covers 0.22–0.41 of the reference mass, against 0.74 for the reference's own top five.

**Few-shot prompting.** With three worked examples in every prompt, the Qwen gains persist or grow (TV −0.057 to −0.087; −0.055 to −0.073 without the true pathology), but OLMo 3.1 32B's list keeps only its higher reference mass (+0.047), with TV, TV without the true pathology and DDF1 no longer significant (Appendix 4.10). Averaging calls narrows the gap: the mean of five

| Pair | Model | TV to reference↓ One list | TV to reference↓ Samples | Reference mass covered One list | Reference mass covered 5 samples | Severe named One list |
|---|---|---|---|---|---|---|
| Qwen3-8B, $T$=0.7 | base | 0.743 | **0.749** | 0.335 | **0.245** | 0.347 |
| | post-trained | **0.700** | 0.793 | **0.396** | 0.207 | **0.437** |
| Qwen3-8B, $T$=1 | base | 0.772 | **0.712** | 0.293 | **0.275** | 0.312 |
| | post-trained | **0.700** | 0.783 | **0.396** | 0.215 | **0.440** |
| Qwen2.5-7B, $T$=0.7 | base | 0.779 | **0.756** | 0.291 | **0.233** | 0.345 |
| | instruct | **0.716** | 0.785 | **0.364** | 0.214 | **0.440** |
| Qwen2.5-7B, $T$=1 | base | 0.803 | **0.715** | 0.255 | **0.257** | 0.304 |
| | instruct | **0.717** | 0.761 | **0.363** | 0.234 | **0.438** |
| OLMo 3 7B, $T$=0.7 | base | 0.795 | **0.757** | 0.268 | **0.237** | 0.256 |
| | SFT+DPO | 0.795 | 0.804 | 0.277 | 0.193 | **0.378** |
| | Instruct (RLVR) | 0.823 | 0.851 | 0.257 | 0.150 | 0.317 |
| | RL-Zero | **0.775** | 0.793 | **0.295** | 0.197 | 0.274 |
| Oracle | reference | 0.265 | 0.18–0.25 | 0.735 | 0.563 | 0.975 |

Table 7: Stated vs. sampled differentials on DDXPlus (980 cases, valid samples only). *One list*: a single sample's stated differential, renormalized over the conditions it names (mean over samples). *Samples*: the empirical distribution of top-1 answers over the 32 samples; *5 samples*: reference mass of the distinct top answers in five random samples, the same five-name budget as one list. *Severe named*: share of (case, severe condition with reference probability $\geq 0.1$) pairs that one list names. Bold marks the better model of each pair (OLMo: of the block). Oracle: the reference's own top five as a list, and draws from the reference as samples. Post-training improves the Qwen lists while worsening their samples; OLMo's lists do not improve.

base-model lists is about as close to the reference as one post-trained list (Qwen3-8B 0.699 against 0.700), so direct-answer post-training mostly packs into one call what the base model spreads across calls; Qwen3-8B's thinking mode goes further, with one list beating five averaged base lists (§4.14).

### 4.7 Consequences for sampling-based methods

We ask what the gap does to two common uses of repeated samples: aggregating them into an answer, and using their agreement as an uncertainty signal (Table 8). The single-call answer is the top answer of a model's first valid sample.

**Aggregation.** Majority voting over 32 samples (self-consistency; 35) raises base models' accuracy by 3.4–5.2 points at $T$=0.7 and 10.4–14.2 at $T$=1. For the general post-trained models it adds at most 0.8 points at $T$=0.7 (Qwen3-8B +0.3, Qwen2.5-7B-Instruct +0.8, OLMo-3-Instruct −0.2; all n.s.) and 0.1–4.4 at $T$=1. Models that reason before answering still benefit (HuatuoGPT-o1 +4.8, m1 +3.8, Qwen3-8B with the CoT prompt +4.5). Averaging the stated lists instead of voting changes accuracy by at most about 2 points for any model. For a top-three differential, the stated list is the better source after post-training. For Qwen3-8B, the top three of one list contains the true pathology 13.7 [+11.4, +15.9] points more often than the three most frequent answers in 32 samples ($T$=1: +10.0), and for Qwen2.5-7B-Instruct 7.3 [+5.3, +9.6] points more often ($T$=1: +2.0, n.s.). For every base model the 32 samples win, by 7.3–13.6 points at $T$=0.7. OLMo-3-Instruct is in between (+1.9, n.s.; $T$=1: −4.7).

**Uncertainty.** We score how well each signal ranks the correctness of the single-call answer (AUROC). The stated probability of that answer needs no extra calls; *agreement*, the share of the model's other samples with the same top answer, uses 4 or 31 more. For base models agreement is far better: 0.80–0.81 with 31 extra samples, against 0.54–0.60 for their nearly uninformative stated probability. Post-training weakens agreement in every family (Qwen3 0.810 → 0.655, Qwen2.5 0.811 → 0.702, OLMo 0.802 → 0.731), consistent with Liu [20], and moves the comparison toward the stated probability. For Qwen3-8B (0.734) and OLMo-3-Instruct (0.796), the stated probability is better than agreement over 32 samples (+0.079 [+0.046, +0.110] and +0.064 [+0.027, +0.101]); at $T$=1 the two are statistically tied. For Qwen2.5-7B-Instruct (0.571 vs. 0.702) and all three medical models, agreement remains better. These AUROCs are computed on each model's own answers, so they compare signals within a model, not models with each other.

**Ambiguity.** The reference differential also says how ambiguous each case is. We correlate each uncertainty signal with the entropy of the case's reference differential. For every model and signal the Spearman correlation lies between −0.10 and +0.14, except AlphaMed's stated probability and stated entropy (+0.30 and +0.32). How uncertain a model is about a case, by either readout, says almost nothing about how ambiguous the case is.

**At 32B.** The same shifts appear with smaller margins. Voting gains fall from 7.0 and 8.4 points for the base models to 1.9 and 0.9 (the latter n.s.) for the Instruct models, and agreement weakens as an error signal (Qwen 0.797 → 0.721, OLMo 0.831 → 0.772). For Qwen2.5-32B-Instruct the top three of one list beats the three most frequent of 32 sampled answers by 5.6 [3.6, 7.7] points, but its stated probability only ties agreement over 32 samples (−0.011, n.s.; it beats agreement over five, +0.047); for OLMo 3.1 32B Instruct agreement stays better (−0.077).

After post-training, then, repeated sampling adds little: voting no longer improves accuracy and agreement carries less signal, while one stated list carries as much of the differential or more. Where the crossover falls depends on the model, so neither readout can be assumed to be the informative one without checking.

### 4.8 Robustness of the stated readout to elicitation

The stated readout above uses a five-item list. We repeat the Qwen pairs and OLMo 3 7B's base and Instruct models at $T$=0.7 with a prompt that asks for a probability on every one of the 49 conditions, summing to one, and take each sample's top answer to be its highest-probability condition (Table 9). The results match the short list. After post-training the stated distribution is closer to the reference (TV −0.052 for Qwen3-8B, −0.067 for Qwen2.5-7B-Instruct, −0.032 for OLMo-3-7B-Instruct) while the samples move away (+0.012 to +0.038). Qwen3-8B's stated probability again beats agreement over 32 samples (+0.069 [+0.033, +0.104]), Qwen2.5-7B-Instruct's again does not (−0.074), and OLMo-3-7B-Instruct's ties it (−0.010; +0.044 against five samples). The top three of one full distribution beats the three most frequent of 32 sampled answers for Qwen3-8B (+14.6 points) but not for the other two models.

### 4.9 Ruling out format, accuracy and floor effects

Table 11 tests simpler explanations of the stated-list result. *Format.* The general models' post-trained lists and base-model lists at $T$=0.7 are nearly all complete (81–99% name five on-list conditions with probabilities; Qwen base models at $T$=1, 61–62%; the medical models, 61–80%). Restricting to complete lists leaves the gaps unchanged (Qwen2.5-7B: 0.779 vs. 0.719). *Accuracy.*

| Model | Single-call accuracy | Vote@32 gain | Top-3: list − 32 samples | AUROC, single-call answer correct | | |
|---|---|---|---|---|---|---|
| | | | | Stated $p$ | Agree@5 | Agree@32 |
| Qwen3-8B-Base | 0.456 | +3.4 | −7.3 | 0.595 | 0.772 | **0.810** |
| Qwen3-8B | 0.556 | +0.3 | +13.7 | **0.734** | 0.613 | 0.655 |
| Qwen3-8B (CoT prompt) | 0.648 | +4.5 | +0.0 | 0.718 | 0.725 | **0.778** |
| Qwen2.5-7B | 0.371 | +5.2 | −8.9 | 0.576 | 0.764 | **0.811** |
| Qwen2.5-7B-Instruct | 0.480 | +0.8 | +7.3 | 0.571 | 0.653 | **0.702** |
| OLMo-3 base | 0.291 | +4.0 | −13.6 | 0.538 | 0.761 | **0.802** |
| OLMo-3 SFT+DPO | 0.327 | −0.4 | −7.8 | 0.738 | 0.702 | **0.751** |
| OLMo-3-Instruct | 0.267 | −0.2 | +1.9 | **0.796** | 0.668 | 0.731 |
| AlphaMed-7B | 0.528 | +0.9 | +0.6 | 0.673 | 0.663 | **0.715** |
| HuatuoGPT-o1-7B | 0.532 | +4.8 | −0.4 | 0.590 | 0.715 | **0.775** |
| m1-7B | 0.495 | +3.8 | −4.6 | 0.680 | 0.736 | **0.782** |

Table 8: Aggregation and uncertainty on DDXPlus ($T$=0.7, 980 cases). Single-call answer: top-1 of the first valid sample. Vote@32 gain: accuracy of the majority top-1 over all valid samples minus single-call accuracy, in points. Top-3: rate at which one stated list's top three contains the true pathology, minus the rate for the three most frequent sampled answers (points). AUROC for the single-call answer being correct: its stated probability (no extra calls) vs. agreement of 4 or 31 further samples; bold marks the best signal. Post-trained models gain little from voting; for some of them the stated probability becomes the better error signal.

A more accurate model puts more stated mass on the true pathology, which the reference also favors. Within samples whose top answer is correct, and within those whose top answer is wrong, the post-trained Qwen list is still closer to the reference (−0.018 to −0.034 and −0.036 to −0.079 in TV across both temperatures; Table 11 shows $T$=0.7). *Floor.* One list can do no better than the reference's own top five, and a finite set of samples no better than as many draws from the reference. Measured as excess over these floors, post-training lowers the list's TV (Qwen3-8B 0.479 → 0.435) and raises the samples' (0.557 → 0.613). *True pathology removed.* With the true pathology removed from both list and reference and both renormalized, the post-trained Qwen and OLMo 3.1 32B lists are still closer to the reference (−0.018 to −0.056), and DDXPlus's set F1 against the reference's support improves by 0.040–0.083 (Table 23). TV, mass and DDF1 are not independent: a list's TV is the mass it leaves out plus the mass it misallocates among the conditions it names (Table 7), and the misallocation rises after post-training in seven of eight settings, so the TV gain comes from the named set. *Patient-blind list.* A fixed list of the five conditions with the highest average reference probability scores TV 0.794, mass 0.238 and DDF1 0.272. Every model's list beats it in TV except OLMo 3 7B's; in DDF1 several base lists do not (Qwen2.5-7B 0.245, OLMo 3 7B 0.225), and the post-trained Qwen and OLMo 3.1 32B lists exceed it by 0.035–0.061. *Over-naming.* As a share of named conditions, post-trained lists name fewer that the reference gives less than 1% (Qwen3-8B 0.59 → 0.55) and fewer unwarranted severe conditions (Qwen2.5-32B 0.74 → 0.65), but as a count per list the change is mixed: from 0.22 fewer unwarranted severe conditions (OLMo 3.1 32B) to 0.35 more (OLMo 3 7B). *Severe conditions one by one.* Table 10 tests each of the 17 severe conditions, counting it wherever the reference gives it any weight. After Benjamini–Hochberg correction, 2–7 conditions per pair are named significantly less often after post-training and 0–6 significantly more often, and no pair names more conditions less often than a random swap of base and post-trained lists would. Significant losses include acute pulmonary edema (Qwen3-8B 0.75 → 0.61, Qwen2.5-32B 0.63 → 0.44, OLMo 3 7B 0.56 → 0.31, OLMo 3.1 32B 0.57 → 0.41), stable angina

| Model | TV to reference↓ One distribution | Samples | AUROC, single-call answer correct Stated $p$ | Agree@5 | Agree@32 | Top-3: one dist. − 32 samples |
|---|---|---|---|---|---|---|
| Qwen3-8B-Base | 0.733 | **0.752** | 0.647 | 0.809 | **0.845** | −6.7 |
| Qwen3-8B | **0.681** | 0.790 | **0.717** | 0.609 | 0.647 | +14.6 |
| Qwen2.5-7B | 0.769 | **0.758** | 0.688 | 0.792 | **0.827** | −10.2 |
| Qwen2.5-7B-Instruct | **0.702** | 0.770 | 0.674 | 0.715 | **0.748** | +1.6 |
| OLMo-3-1025-7B | 0.807 | **0.766** | 0.610 | 0.744 | **0.786** | −15.3 |
| OLMo-3-7B-Instruct | **0.775** | 0.787 | 0.749 | 0.705 | **0.759** | −9.5 |

Table 9: Full-distribution prompt on DDXPlus ($T$=0.7, 980 cases): a probability on every condition, top answer = highest stated probability. TV: one sample's stated distribution vs. the sampled top-1 answers (bold: better model of each pair). AUROC: bold marks each model's best signal.

| Pair (base → post) | Named less / more often (point estimates) | Significant (FDR 5%) less / more | Expected less often by chance | $P$(at least as many less) |
|---|---|---|---|---|
| Qwen3-8B, $T$=0.7 | 5 / 12 | 2 / 3 | 8.5 | 0.98 |
| Qwen3-8B, $T$=1 | 6 / 11 | 2 / 4 | 8.5 | 0.94 |
| Qwen2.5-7B, $T$=0.7 | 5 / 12 | 2 / 2 | 8.6 | 0.99 |
| Qwen2.5-7B, $T$=1 | 4 / 13 | 2 / 5 | 8.6 | 0.99 |
| Qwen2.5-32B, $T$=0.7 | 6 / 11 | 3 / 5 | 8.5 | 0.93 |
| OLMo 3 7B, $T$=0.7 | 8 / 9 | 7 / 6 | 8.5 | 0.69 |
| OLMo 3 7B, $T$=1 | 8 / 9 | 6 / 6 | 8.5 | 0.70 |
| OLMo 3.1 32B, $T$=0.7 | 7 / 10 | 3 / 0 | 8.5 | 0.83 |

Table 10: The 17 severe conditions one by one (DDXPlus). For each condition, the share of a model's lists that name it, over the cases whose reference gives it any weight; differences post-trained minus base with intervals that resample the true pathologies (10,000 resamples) and Benjamini–Hochberg correction across the 17 conditions. The last two columns compare the number of conditions named less often with a null that swaps each case's base and post-trained lists at random (2,000 swaps).

(Qwen2.5-7B 0.25 → 0.08), NSTEMI/STEMI (OLMo 3 7B 0.31 → 0.14, OLMo 3.1 32B 0.49 → 0.36) and myocarditis (Qwen2.5-32B 0.14 → 0.05); significant gains include pulmonary embolism (Qwen2.5-7B 0.29 → 0.56, OLMo 3 7B 0.20 → 0.65), NSTEMI/STEMI (Qwen2.5-7B 0.32 → 0.52) and unstable angina (Qwen3-8B 0.14 → 0.37). Among single stages, OLMo 3 7B's RLVR step and OLMo 3.1 32B's SFT and RLVR steps do name more conditions less often than chance (12–13 of 17, $p$=0.03–0.04).

*Sampling from the list.* Replacing each sample's top answer with a draw from its own stated list raises pass@32 for every model. Compared like for like, post-trained lists reach higher than their base model's lists in five of seven comparisons (Qwen3-8B 0.811 vs. 0.780 at $T$=0.7, Qwen2.5-32B-Instruct 0.880 vs. 0.850, OLMo 3.1 32B Instruct 0.853 vs. 0.839), but not for Qwen3-8B at $T$=1 (0.820 vs. 0.832) or OLMo 3 7B (0.577 vs. 0.719), and a uniform guesser reaches 0.48. OLMo 3 7B remains the exception throughout: its Instruct list is closer to the reference when its top answer is correct but much farther when it is wrong.

### 4.10 Few-shot elicitation

The main analyses give every model the same zero-shot chat prompt, which base models were not trained to follow; they give invalid answers on up to 47% of samples, which we count as wrong.

| Model ($T$=0.7) | Complete lists | TV of one list↓ complete only | top-1 right | top-1 wrong | Excess TV over floor↓ One list | Samples | Severe named ref. $< 0.1$ | pass@32 list-sampled |
|---|---|---|---|---|---|---|---|---|
| Qwen3-8B-Base | 0.90 | 0.743 | 0.663 | 0.806 | 0.479 | **0.557** | 0.752 | 0.780 |
| Qwen3-8B | 0.99 | **0.701** | **0.646** | **0.769** | **0.435** | 0.613 | **0.680** | **0.811** |
| Qwen2.5-7B | 0.95 | 0.779 | 0.689 | 0.828 | 0.514 | **0.575** | 0.823 | 0.772 |
| Qwen2.5-7B-Instruct | 0.91 | **0.719** | **0.655** | **0.773** | **0.452** | 0.605 | **0.796** | **0.802** |
| Qwen2.5-32B | 0.96 | 0.725 | 0.661 | 0.795 | 0.462 | **0.543** | 0.735 | 0.850 |
| Qwen2.5-32B-Instruct | 0.98 | **0.674** | **0.639** | **0.726** | **0.409** | 0.579 | **0.645** | **0.880** |
| OLMo-3-1025-7B | 0.81 | **0.794** | 0.687 | **0.837** | **0.524** | **0.522** | **0.747** | **0.719** |
| OLMo-3-7B-Instruct | 0.95 | 0.823 | **0.659** | 0.882 | 0.558 | 0.671 | 0.779 | 0.577 |
| OLMo-3-1125-32B | 0.89 | 0.744 | 0.662 | 0.812 | 0.481 | **0.532** | 0.706 | 0.839 |
| OLMo-3.1-32B-Instruct | 0.99 | **0.701** | **0.644** | **0.774** | **0.436** | 0.583 | **0.634** | **0.853** |

Table 11: Robustness of the stated-list result (DDXPlus, $T$=0.7, 980 cases, valid samples). Complete lists: share naming five on-list conditions with valid probabilities. TV of one list on complete lists only, and on samples whose top answer is right or wrong. Excess TV: TV minus the expected TV of the matching oracle (the reference's top five for one list; as many reference draws as the model has valid samples for the samples). Severe named, ref. $< 0.1$: share of named severe conditions that the reference gives less than 0.1 (over-naming). pass@32 list-sampled: top answers drawn from each sample's stated list. Bold: better model of each pair. Excess TV is averaged over samples against each case's floor, so it is not exactly the difference of the mean TV and the mean floor. A uniform guesser scores about 0.48 at pass@32.

We therefore re-ran the five final base/post-trained pairs at $T$=0.7 with three worked examples in the prompt: three fixed DDXPlus training patients with different true pathologies, each followed by the top five conditions of its reference differential, with probabilities, in the JSON format the instructions ask for. Base models see the instructions and examples as a plain document and complete the next one (generation stops at the end of the list); post-trained models see the same instructions and examples as earlier chat turns. Invalid samples fall below 0.5% for every model (Table 12).

The examples help base models most. Their pass@1 rises from 0.178 to 0.306 for OLMo-3-1025-7B, from 0.326 to 0.528 for OLMo-3-1125-32B and from 0.500 to 0.647 for Qwen2.5-32B, and their pass@32 reaches 0.76–0.90, while their stated lists barely change (TV 0.712–0.787, against 0.726–0.794 zero-shot). The losses in the samples therefore grow (Table 13): post-training costs 15.2–29.9 points of pass@32 in every pair, including both 32B pairs, whose zero-shot losses did not survive correction, and every post-trained model's Cov@32 (0.21–0.25) is below every base model's (0.33–0.45) and below a uniform guesser's.

For the three Qwen pairs the stated-list improvement persists and mostly grows: TV $-0.057$ to $-0.087$, $-0.055$ to $-0.073$ with the true pathology removed, and DDF1 $+0.051$ to $+0.081$, all surviving correction. OLMo 3 7B remains the exception (TV $+0.021$, n.s.; $+0.024$ without the true pathology). For OLMo 3.1 32B the improvement shrinks: the post-trained list still places more reference mass on the conditions it names ($+0.047$), but its TV ($-0.022$), TV without the true pathology ($-0.006$) and DDF1 ($+0.009$) no longer survive correction. At 32B the stated-list result is therefore robust for Qwen2.5-32B but depends on the prompt for OLMo 3.1 32B. Post-trained lists still name 5–10 of the 17 severe conditions less often than their base model's.

The consequences replicate in all five pairs. Majority voting over 32 samples gains base models 6.2–10.9 points and post-trained models 0.2–1.1. The three most frequent of 32 sampled answers beat the top three of one stated list by 8.2–22.0 points for base models, while for post-trained models the list is ahead by 0.2–16.7. Agreement with four more samples is the better error signal for every base model (AUROC 0.79–0.84, against 0.67–0.73 for the stated probability), and the

| Model | Invalid (%) | pass@1 | pass@32 | Cov@32 | TV, one list all | TV, one list w/o true | DDF1 | TV samples | Vote@32 gain | Top-3 list − samples | AUROC stated | AUROC agree@5 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Qwen3-8B-Base | 0.1 | 0.403 | **0.809** | **0.383** | 0.768 | 0.860 | 0.245 | **0.732** | +8.4 | −16.0 | 0.722 | 0.828 |
| Qwen3-8B | 0.0 | **0.601** | 0.639 | 0.221 | **0.682** | **0.805** | **0.327** | 0.786 | +0.2 | +14.5 | 0.693 | 0.563 |
| Qwen2.5-7B (base) | 0.4 | 0.398 | **0.761** | **0.333** | 0.784 | 0.879 | 0.238 | **0.756** | +6.2 | −15.6 | 0.734 | 0.792 |
| Qwen2.5-7B-Instruct | 0.0 | **0.450** | 0.535 | 0.220 | **0.702** | **0.806** | **0.309** | 0.798 | +0.6 | +16.7 | 0.757 | 0.594 |
| Qwen2.5-32B (base) | 0.1 | 0.647 | **0.896** | **0.343** | 0.712 | 0.847 | 0.298 | **0.715** | +7.1 | −8.2 | 0.726 | 0.838 |
| Qwen2.5-32B-Instruct | 0.0 | **0.663** | 0.744 | 0.252 | **0.654** | **0.775** | **0.349** | 0.758 | +0.6 | +10.1 | 0.718 | 0.647 |
| OLMo-3-1025-7B (base) | 0.2 | 0.306 | **0.857** | **0.454** | **0.787** | **0.859** | **0.245** | **0.712** | +10.9 | −22.0 | 0.674 | 0.823 |
| OLMo-3-7B-Instruct | 0.1 | **0.338** | 0.558 | 0.213 | 0.808 | 0.884 | 0.238 | 0.830 | +0.7 | +0.2 | 0.733 | 0.668 |
| OLMo-3-1125-32B (base) | 0.4 | 0.528 | **0.888** | **0.397** | 0.719 | 0.818 | 0.302 | **0.699** | +8.9 | −13.4 | 0.670 | 0.826 |
| OLMo-3.1-32B-Instruct | 0.0 | **0.575** | 0.697 | 0.249 | **0.697** | **0.812** | **0.311** | 0.772 | +1.1 | +3.7 | 0.772 | 0.684 |

Table 12: Few-shot elicitation on DDXPlus ($T$=0.7, 980 cases, 32 samples; three worked examples in the prompt for every model, §4.10). Invalid: samples naming no listed condition. TV: distance to the reference differential, for one stated list (with the true pathology removed from both in "w/o true") and for the sampled top answers. Vote@32 gain and top-3 list minus top-3 of the samples are in accuracy points. AUROC: for the single-call answer being correct, of its stated probability and of agreement with the next 4 samples. Bold: better of the pair.

| Pair (base → post) | pass@32 (points) | TV samples | TV, one list all | TV, one list w/o true | DDF1 | Reference mass of one list |
|---|---|---|---|---|---|---|
| Qwen3-8B | −12.4 / −17.0 | +0.043 / +0.054 | −0.043 / −0.087 | −0.018$^{\dagger}$ / −0.055 | +0.042 / +0.081 | +0.062 / +0.113 |
| Qwen2.5-7B | −9.7 / −22.7 | +0.028$^{\dagger}$ / +0.042 | −0.062 / −0.082 | −0.041 / −0.073 | +0.062 / +0.072 | +0.073 / +0.124 |
| Qwen2.5-32B | −4.9$^{\dagger}$ / −15.2 | +0.033 / +0.044 | −0.053 / −0.057 | −0.043 / −0.072 | +0.051 / +0.051 | +0.067 / +0.084 |
| OLMo 3 7B | −28.3 / −29.9 | +0.090 / +0.118 | +0.029$^{\dagger}$ / +0.021$^{\dagger}$ | +0.034 / +0.024 | −0.006$^{\dagger}$ / −0.007$^{\dagger}$ | −0.010$^{\dagger}$ / −0.002$^{\dagger}$ |
| OLMo 3.1 32B | −4.0$^{\dagger}$ / −19.1 | +0.020$^{\dagger}$ / +0.074 | −0.045 / −0.022$^{\dagger}$ | −0.023 / −0.006$^{\dagger}$ | +0.040 / +0.009$^{\dagger}$ | +0.056 / +0.047 |

Table 13: Base → post-trained differences under the zero-shot prompt / the few-shot prompt (DDXPlus, $T$=0.7). $^{\dagger}$: not significant under the condition-clustered bootstrap with Holm correction (over 112 tests for the zero-shot and 70 for the few-shot pairs; Appendix 6).

stated probability for every post-trained model (0.69–0.77, against 0.56–0.68). This check uses one temperature and one set of examples, and the examples show the reference's own lists, which may help every model match the reference.

### 4.11 Two further controls

**Reference-free few-shot examples.** The examples of §4.10 show each training patient's reference top five, and so also the target of the evaluation. We re-ran the six final pairs with the same three patients but with each example list replaced by Qwen2.5-32B-Instruct's own greedy answer (on-list, with the true pathology first for two patients and second for one). The conclusions do not change (Table 14). Post-training costs 12.7–29.7 points of pass@32 in every pair. The stated list moves closer to the reference for the three Qwen pairs and, now clearly, for Mistral (TV −0.053 to −0.077; −0.031 to −0.059 with the true pathology removed), but not for OLMo 3 7B or OLMo 3.1 32B, and the misallocation among named conditions rises wherever the list improves. Voting helps base models by 3.7–13.7 points and post-trained ones by at most 1.4; the top three of one list beats the three most frequent samples for every post-trained model; the stated probability beats agreement as an error signal for every post-trained model and agreement beats it for every base model; and no pair names more severe conditions less often than chance. Base models score lower at $k$=1 than with reference-derived examples (OLMo-3-1125-32B 0.453 against 0.528).

| Pair (base → post) | pass@32 (points) | TV samples | TV, one list all | TV, one list w/o true | DDF1 | Misallocated |
|---|---|---|---|---|---|---|
| Qwen3-8B | −14.6 | +0.049 | −0.077 | −0.049 | +0.079 | +0.016 |
| Qwen2.5-7B | −19.7 | +0.031† | −0.077 | −0.059 | +0.074 | +0.024 |
| Qwen2.5-32B | −14.5 | +0.036 | −0.053 | −0.057 | +0.054 | +0.013 |
| OLMo 3 7B | −29.7 | +0.116 | −0.007† | +0.013† | −0.002† | +0.002† |
| OLMo 3.1 32B | −17.4 | +0.073 | −0.016† | +0.002† | +0.009† | +0.006† |
| Mistral-7B | −12.7 | +0.038† | −0.059 | −0.031 | +0.057 | +0.016 |

Table 14: Few-shot control with reference-free examples (DDXPlus, $T$=0.7, 980 cases): the same three training patients as in §4.10, but each example list is Qwen2.5-32B-Instruct's own greedy answer rather than the reference's top five. Post-trained minus base. †: not significant (condition-clustered bootstrap, Holm over 84 tests; for the misallocation, the clustered interval includes zero).

| Model | $T$ | Invalid (%) | pass@1 | pass@32 | Distinct | TV samples | TV one list | Vote@32 gain | AUROC stated | AUROC agree@32 |
|---|---|---|---|---|---|---|---|---|---|---|
| Qwen3-8B-Base | 0.7 | 11.3 | 0.388 | 0.732 | 3.77 | 0.750 | 0.743 | +3.4 | 0.595 | 0.810 |
| | 1 | 22.3 | 0.282 | 0.789 | 6.60 | 0.715 | 0.775 | +10.4 | 0.597 | 0.857 |
| Qwen3-8B | 0.7 | 0.0 | 0.559 | 0.607 | 1.27 | 0.793 | 0.700 | +0.3 | 0.734 | 0.655 |
| | 1 | 0.0 | 0.556 | 0.643 | 1.53 | 0.783 | 0.700 | +1.4 | 0.726 | 0.701 |
| | 1.3 | 0.0 | 0.552 | 0.668 | 1.73 | 0.776 | 0.701 | +1.0 | 0.707 | 0.724 |
| | 1.6 | 0.0 | 0.549 | 0.684 | 1.92 | 0.770 | 0.701 | +1.8 | 0.715 | 0.745 |
| Qwen2.5-7B | 0.7 | 0.9 | 0.352 | 0.695 | 4.91 | 0.757 | 0.779 | +5.2 | 0.576 | 0.811 |
| | 1 | 18.4 | 0.239 | 0.766 | 8.23 | 0.719 | 0.805 | +14.1 | 0.587 | 0.867 |
| Qwen2.5-7B-Instruct | 0.7 | 0.0 | 0.478 | 0.598 | 1.82 | 0.785 | 0.717 | +0.8 | 0.571 | 0.702 |
| | 1 | 0.0 | 0.464 | 0.663 | 2.65 | 0.761 | 0.717 | +4.4 | 0.583 | 0.767 |
| | 1.3 | 2.2 | 0.436 | 0.704 | 3.38 | 0.745 | 0.722 | +5.6 | 0.602 | 0.794 |
| | 1.6 | 30.5 | 0.304 | 0.726 | 3.88 | 0.729 | 0.749 | +10.0 | 0.637 | 0.814 |

Table 15: Matched-temperature control (DDXPlus, 980 cases, 32 samples, zero-shot prompt): the post-trained models sampled at higher temperatures (top-$p$ 1) against their base models at $T$=0.7 and $T$=1. Columns as in Tables **??**, 7 and 8.

**Matched temperature.** Post-trained models are sharper than their base models at the same nominal temperature, so part of the narrowing could be colder effective sampling. We sampled Qwen3-8B and Qwen2.5-7B-Instruct at $T$=1.3 and 1.6 (Table 15), and the two models differ. Qwen3-8B barely broadens: at $T$=1.6 it gives 1.92 distinct answers and reaches 0.684 at pass@32, still below its base model at $T$=0.7 (0.732) and at $T$=1 (0.789; −10.5 points, surviving correction); voting gains stay under 2 points, agreement stays weaker than for the base model, and its stated list does not change (TV 0.701). Qwen2.5-7B-Instruct does broaden: at $T$=1.3, with 2% invalid answers, it gives 3.38 distinct answers, matches its base model's pass@32 at $T$=0.7 (0.704 against 0.695), gains 5.6 points from voting, and agreement is again the better error signal (0.794 against 0.602), at a cost of 4.2 points of pass@1; at $T$=1.6, 31% of its answers are invalid. Its stated list stays closer to the reference than the base model's (0.722 against 0.779). For some models, then, much of the sampled narrowing is sharper sampling that a higher temperature undoes; the stated-list result is not.

### 4.12 A third family: Mistral-7B v0.3

We repeat the comparison with Mistral-7B-v0.3 and Mistral-7B-Instruct-v0.3, a family also studied by Jang et al. [11]. With the zero-shot prompt the base model does not answer: 52% of its samples at $T$=0.7 and 76% at $T$=1 are invalid (it continues the document instead), and its valid answers mostly repeat the first listed condition, so its single-call accuracy is 6% and the zero-shot pair is not informative. With the few-shot prompt of §4.10 the base model answers validly (0.4% invalid), and the pair reproduces the other families (Table 16). Post-training raises pass@1 by 14.0 points and lowers pass@32 by 14.5, cuts distinct answers from 5.0 to 1.3 and Cov@32 by 0.138, and moves the samples away from the reference (TV +0.058). Voting gains the base model 5.6 points and the Instruct model 0.8; the three most frequent of 32 samples beat one list by 12.6 points for the base model, while for the Instruct model the list is ahead by 7.0; agreement is the better error signal for the base model (AUROC 0.826 against 0.671) and the stated probability for the Instruct model (0.747 against 0.613). The stated list improves modestly (TV −0.036 and DDF1 +0.046, surviving clustering and Holm correction over 42 tests; −0.023 with the true pathology removed, which does not), the smallest gain of the three families, and 11 of the 17 severe conditions are named less often. Mistral-Instruct also names 0.4 conditions per list that are not among the 49.

### 4.13 Thinking mode

Qwen3-8B can reason in a thinking block before it answers; the main runs turn this off. With thinking on (zero-shot prompt, $T$=0.7, up to 8,192 tokens; traces average about 1,500 tokens and all end before the limit), the same weights behave differently (Table 16). Accuracy rises at both ends: pass@1 is 0.671 against 0.559 without thinking, and pass@32 is 0.815, above both the non-thinking model (0.607) and the base model (0.732; +8.4 [1.7, 15.3] points, not surviving Holm correction over 42 tests). The samples are still narrow (1.76 distinct answers against 3.77 for the base model; Cov@32 0.283 against 0.323) but no farther from the reference than the base model's (TV 0.745 against 0.750). The stated list is the closest of any 7–8B model (TV 0.670, against 0.700 without thinking). The shift in error signals reverses: agreement is again the better signal (AUROC 0.693 with five samples, against 0.638 for the stated probability), as for base models, and voting gains 1.5 points (n.s.). The narrowing of the samples and the advantage of the stated probability are therefore properties of answering directly, which a reasoning mode largely undoes, consistent with the chain-of-thought prompt of §4.3.

### 4.14 Averaging stated lists across calls

Stated lists can be averaged across calls, as samples can be voted. For each case we average the first 5 or 32 valid stated lists, each renormalized over the conditions it names, and score the mean against the reference (Table 17). Averaging helps base models far more than post-trained ones, whose lists vary little between calls (with the zero-shot prompt, 32 averaged lists name 7.6–13.4 conditions, against 16.6–28.8 for base models). The mean of five base-model lists is about as close to the reference as one post-trained list (Qwen3-8B 0.699 against 0.700, Qwen2.5-7B 0.717 against 0.717, OLMo 3.1 32B 0.691 against 0.701). With five lists on both sides, the post-trained mean is closer for Qwen2.5-7B under both prompts (−0.025 and −0.027) and for Qwen2.5-32B and Qwen3-8B under one prompt each, not significantly closer for OLMo 3.1 32B (zero-shot) and Mistral (few-shot), and farther for OLMo 3 7B and few-shot OLMo 3.1 32B. Averaged base lists also name severe conditions at least as often, and more often for OLMo 3.1 32B, OLMo 3 7B, Mistral and few-shot Qwen2.5-32B (by 0.06–0.14). Direct-answer post-training thus mostly packs into one call the differential that the base model spreads across calls, as Lake et al. [16] find for opinions, rather

| Model | Invalid (%) | pass@1 | pass@32 | Distinct | Cov@32 | TV, one list all | TV, one list w/o true | DDF1 | TV samples | Vote@32 gain | Top-3 list − samples | AUROC stated | AUROC agree@5 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Mistral-7B-v0.3 (base), few-shot | 0.4 | 0.428 | **0.791** | **5.02** | **0.355** | 0.777 | 0.875 | 0.250 | **0.741** | +5.6 | −12.6 | 0.671 | 0.826 |
| Mistral-7B-Instruct-v0.3, few-shot | 0.3 | **0.568** | 0.646 | 1.34 | 0.217 | **0.741** | **0.851** | **0.296** | 0.799 | +0.8 | +7.0 | 0.747 | 0.613 |
| Qwen3-8B-Base | 11.3 | 0.388 | 0.732 | **3.77** | **0.323** | 0.743 | 0.833 | 0.276 | 0.750 | +3.4 | −7.3 | 0.595 | 0.772 |
| Qwen3-8B, non-thinking | 0.0 | 0.559 | 0.607 | 1.27 | 0.217 | 0.700 | 0.815 | 0.318 | 0.793 | +0.3 | +13.7 | 0.734 | 0.613 |
| Qwen3-8B, thinking | 0.0 | **0.671** | **0.815** | 1.76 | 0.283 | **0.670** | **0.799** | **0.349** | **0.745** | +1.5 | +3.0 | 0.638 | 0.693 |

Table 16: A third family and a reasoning mode on DDXPlus ($T$=0.7, 980 cases, 32 samples). Top: Mistral-7B v0.3 with the few-shot prompt of §4.10. Bottom: Qwen3-8B-Base and Qwen3-8B with thinking off (the main runs) and on, with the zero-shot prompt. Columns as in Table 12; Distinct: distinct top answers per 32 samples. Bold: best within each block.

| Pair | TV, base model 1 list | TV, base model mean of 5 | TV, base model mean of 32 | TV, post-trained model 1 list | TV, post-trained model mean of 5 | TV, post-trained model mean of 32 | Post − base, 5 lists each TV | Post − base, 5 lists each Severe named |
|---|---|---|---|---|---|---|---|---|
| *Zero-shot prompt* | | | | | | | | |
| Qwen3-8B | 0.743 | 0.699 | 0.691 | 0.700 | 0.695 | 0.692 | −0.004† | +0.030† |
| Qwen2.5-7B | 0.779 | 0.717 | 0.706 | 0.717 | 0.692 | 0.685 | −0.025 | +0.023† |
| Qwen2.5-32B | 0.726 | 0.681 | 0.674 | 0.674 | 0.658 | 0.654 | −0.024 | −0.037† |
| OLMo 3 7B | 0.794 | 0.730 | 0.717 | 0.823 | 0.805 | 0.801 | +0.074 | −0.076 |
| OLMo 3.1 32B | 0.746 | 0.691 | 0.678 | 0.701 | 0.677 | 0.673 | −0.014† | −0.088 |
| *Few-shot prompt* | | | | | | | | |
| Qwen3-8B | 0.768 | 0.704 | 0.689 | 0.682 | 0.673 | 0.672 | −0.030 | −0.006† |
| Qwen2.5-7B | 0.784 | 0.712 | 0.694 | 0.702 | 0.686 | 0.683 | −0.027 | +0.012† |
| Qwen2.5-32B | 0.712 | 0.655 | 0.641 | 0.654 | 0.640 | 0.638 | −0.015† | −0.094 |
| OLMo 3 7B | 0.787 | 0.703 | 0.676 | 0.808 | 0.786 | 0.778 | +0.083 | −0.141 |
| OLMo 3.1 32B | 0.719 | 0.651 | 0.630 | 0.697 | 0.682 | 0.680 | +0.031 | −0.060 |
| Mistral-7B v0.3 | 0.777 | 0.708 | 0.691 | 0.741 | 0.723 | 0.720 | +0.015† | −0.118 |
| *Zero-shot prompt, thinking on* | | | | | | | | |
| Qwen3-8B, thinking | 0.743 | 0.699 | 0.691 | 0.670 | 0.650 | 0.647 | −0.049 | +0.129 |

Table 17: Averaging stated lists across calls (DDXPlus, $T$=0.7, 980 cases). TV to the reference differential of one stated list (per-case mean over valid lists, as in Table 7) and of the mean of the first 5 or 32 valid lists, for the base and the final post-trained model. Last two columns: post-trained minus base with five averaged lists each, for TV and for the share of severe conditions with reference probability $\geq 0.1$ that are named. †: the condition-clustered 95% interval includes zero.

than adding information that averaged base lists lack. Qwen3-8B's thinking mode goes furthest: five of its lists beat five base-model lists by more than any direct-answer model's (TV −0.049) and name more severe conditions (+0.129). For a post-trained model answering directly, averaging five lists instead of reading one improves TV by only 0.005–0.025 at $T$=0.7 (up to 0.038 at $T$=1); it names 6–17 points more severe conditions, mostly by naming one to three more conditions. Five post-trained lists still cover 0.44–0.50 of the reference mass at $T$=0.7 (OLMo 3 7B 0.37) with six to eight conditions, about twice what their 32 samples cover (0.22–0.28) and as much as a uniform guesser naming about 24 conditions (0.48), so the coverage deficit of §**??** is one of the sampled readout.

### 4.15 Which signal to use: combination and held-out audit

Whether the stated probability or agreement across samples is the better error signal varies across models, so we ask what predicts it and whether a practitioner can tell. We pool 56 model–setting combinations scored on DDXPlus: the base/post-trained pairs at both temperatures and with three

prompts, the OLMo stages, the 32B models, the medical models, our GRPO runs, and the full-distribution prompt. They share weights, prompts and temperatures, so we treat the correlations below as descriptive. The stated probability's advantage over five-sample agreement correlates with the *single-answer rate*, the share of cases whose five samples all give the same top answer (Spearman 0.75), with the stated probability's own AUROC (0.82; the gap contains this term, so part of this is by construction), and negatively with the number of distinct answers (−0.74). The stated probability is the better signal for 19 of the 56. No single-answer threshold separates them: Qwen2.5-7B-Instruct still favors agreement, while OLMo 3 7B's DPO and Instruct models favor their unusually informative stated probability. Combining the signals beats choosing between them. Fitting a logistic model of correctness on the stated probability and five-sample agreement with 100 labeled cases, and evaluating on the other 880, gives a mean held-out AUROC of 0.753 for the 39 post-trained combinations (0.765 for all 56); averaging the two signals' ranks without fitting gives 0.746; picking the better single signal on the same 100 cases gives 0.720; always using agreement 0.695; always using the stated probability 0.671. The logistic combination beats the single-signal pick for 35 of the 39 post-trained combinations. This matches the general finding that hybrids of consistency and verbalized confidence beat either [38]. For choosing a single signal, the simplest rule, always using agreement, is right for 66% of the combinations (54% of the 39 post-trained ones) and loses 0.025 AUROC on average (0.032). An audit does better: choose the signal with the higher AUROC on 100 random labeled cases, using five samples per case, and evaluate the choice on the other 880. Over 500 draws per combination it picks the better held-out signal 86% of the time (82% for post-trained models) and loses 0.005 AUROC (0.007); with 50 cases, 82% and 0.010; with 200, 90% and 0.003. It is reliable when the two signals differ by more than 0.05 AUROC (96% of draws, 39 combinations) and close to a coin flip when they differ by less than 0.02 (53%, 5 combinations), where the choice matters least. The audit's main cost is gold labels for 100 cases from the target population, and we validate it only within DDXPlus.

### 4.16 Medical post-training

Applied after instruction tuning, medical RL does not cost coverage. AlphaMed improves on its parent at every $k$ (Figure 4): +3.9 points at $k$=1 ([+2.2, +5.5]; clustered [−0.4, +8.4]), widening to +8.4 at $k$=32 ([+6.0, +10.6]; clustered [+2.6, +14.7]). It reaches at least one correct sample for 47 of 49 conditions, against 39 for its parent. It does not restore breadth, though: distinct answers (1.87 vs. 1.82), entropy (0.28 vs. 0.30), and coverage (0.27 vs. 0.24) stay at the instruction-tuned level, far below the base model. AlphaMed also names off-list conditions more often than its parent, a mild loss of instruction following. By condition, AlphaMed slightly *raises* consideration in every severity tier (+2.2 to +4.3 points; Table 3). It still loses acute pulmonary edema (0.31 to 0.13), and it introduces an attractor of its own: spontaneous pneumothorax as the top answer in 10.3% of samples, against 4.3% for its parent and a reference rate of 1.0%. Medical post-training thus operates on an already collapsed distribution; the collapse comes from the general post-training stage. The other two descendants go further (Table 18). HuatuoGPT-o1 (SFT then PPO) and m1 (SFT distillation from a reasoning teacher) both *widen* the differential relative to their parent: 2.49 and 3.29 distinct answers vs. 1.82, and coverage@32 of 0.31 and 0.33 vs. 0.24. Their pass@32 rises by 13.7 [+11.3, +16.0] and 18.1 [+15.5, +20.7] points. m1 gains almost nothing at $k$=1 (+0.9, interval includes zero), so its improvement lies entirely in breadth. Both reasoning-trained models are also less calibrated (ECE 0.17 and 0.21 vs. 0.11). Medical post-training on an instruction-tuned model thus never adds to the collapse, and reasoning distillation partially reverses it.

| Model | Post-training | pass@1 | pass@32 | Distinct | Entropy | Cov@32 | ECE |
|---|---|---|---|---|---|---|---|
| Qwen2.5-7B-Instruct | (parent) | 0.478 | 0.598 | 1.82 | 0.30 | 0.239 | **0.106** |
| AlphaMed-7B | rule-based RL | 0.516 | 0.682 | 1.87 | 0.28 | 0.266 | 0.121 |
| HuatuoGPT-o1-7B | SFT + PPO | **0.536** | 0.735 | 2.49 | 0.44 | 0.306 | 0.168 |
| m1-7B-23K | SFT distillation | 0.486 | 0.779 | 3.29 | 0.62 | 0.328 | 0.206 |
| Qwen2.5-7B (base) | (parent) | 0.352 | 0.695 | **4.91** | **1.00** | **0.335** | 0.291 |
| AlphaMed-7B (base-RL) | rule-based RL | 0.495 | **0.790** | 3.14 | 0.62 | 0.320 | 0.202 |

Table 18: Medical models on DDXPlus (980 cases, 32 samples, $T$=0.7). Upper block: descendants of Qwen2.5-7B-Instruct; none narrows the differential further, and the reasoning-trained models widen it. Lower block: AlphaMed's companion checkpoint trained from the base model, which keeps most of the base model's diversity and has the best zero-shot pass@32 of any 7B model.

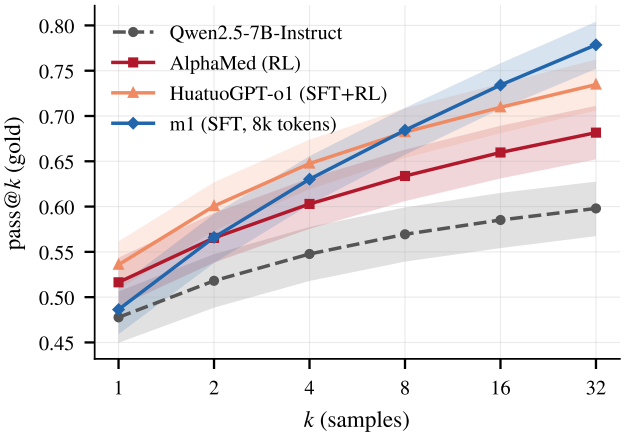



Figure 4: DDXPlus at $T$=0.7 (980 cases): Qwen2.5-7B-Instruct (dashed) vs. its medical descendants, with 95% case-bootstrap bands. Every descendant matches or beats its parent at every $k$, and the gaps widen with $k$: there is no sharpening tax at the medical stage.

**Medical RL from the base model.** Alongside the instruct-initialized AlphaMed-7B, the AlphaMed authors released a companion checkpoint, AlphaMed-7B-base-rl, initialized from Qwen2.5-7B rather than Qwen2.5-7B-Instruct (according to its configuration file). The paper [19] describes only the instruct-initialized models: GRPO with a binary correctness reward on about 19k multiple-choice questions from MedQA and MedMCQA, with a 64-question by 8-sample batch for 300 steps. It does not document the base-initialized checkpoint, so we cannot rule out differences in data or hyperparameters, and we treat it as corroborating rather than controlled evidence. It offers a third-party check on whether RL itself causes the collapse (Table 18, lower block). It does not. RL from the base model improves on the base at every $k$ (+14.3 [+13.1, +15.7] points at $k$=1, +9.5 [+7.4, +11.6] at $k$=32). It keeps 3.14 distinct answers, against 1.87 for the instruct-initialized AlphaMed. Against that instruct-model AlphaMed, it is 2.2 [0.4, 3.9] points worse at $k$=1 and 10.8 [+8.6, +13.2] points better at $k$=32. Its severe-tier Considered@32 shows no detectable change relative to the base (−0.7 [−3.0, +1.7] points), against −12.4 for instruction tuning and −8.1 for instruction tuning followed by AlphaMed's RL (Table 3; all well below the chance levels of §4.2). RL from a base model also narrows less in OLMo's RL-Zero at $T$=1 and in our GRPO runs (§5), but not uniformly: RL-Zero at $T$=0.7 loses about as much as SFT, and within OLMo 3 7B's instruct pipeline the RL stage after SFT and DPO is the largest single narrowing (§4.4).

| Model | ≤2022 ($n$=901) pass@1 | pass@32 | 2023–24 ($n$=348) pass@1 | pass@32 | 2025+ ($n$=152) pass@1 | pass@32 |
|---|---|---|---|---|---|---|
| Qwen2.5-7B (base) | 2.0 | 8.2 | 0.0 | 0.3 | 0.6 | 3.3 |
| Qwen2.5-7B-Instruct | 2.3 | 7.7 | 0.0 | 0.9 | 0.8 | 3.9 |
| AlphaMed-7B | 3.0 | 9.0 | 0.3 | 1.4 | 1.4 | 3.9 |
| AlphaMed-7B (base-RL) | 2.4 | 8.8 | 0.1 | 1.1 | 1.4 | 4.6 |
| HuatuoGPT-o1-7B | 3.1 | 8.4 | 0.3 | 1.4 | 1.3 | 3.9 |
| m1-7B | 2.6 | 9.5 | 0.3 | 1.7 | 1.0 | 3.9 |
| Qwen3-8B-Base | 2.1 | 8.3 | 0.1 | 0.6 | 1.1 | 3.9 |
| Qwen3-8B | 3.1 | 5.2 | 0.1 | 0.3 | 2.0 | 2.6 |

Table 19: Phenopacket Store accuracy (%, exact diagnosis) by publication year of the source case report (1,401 cases, 32 samples, $T$=0.7). Every model is far more accurate on cases published up to 2022 than on 2023–24 cases; 2025+ cases fall in between.

### 4.17 Rare disease: overconfidence without accuracy

On 1,401 Phenopacket Store cases covering 749 diseases, every 7–8B model is near the floor: pass@1 is 1.4–2.2% and pass@32 3.7–7.0% for the Qwen-family models, and lower for OLMo 3 7B (2.5–4.5%). Stated confidence is not near the floor. Qwen2.5-7B-Instruct states a mean top-1 probability of 0.51, and its RL-trained medical descendants state 0.67 (HuatuoGPT-o1) and 0.69 (AlphaMed), while their accuracy is about 2%. ECE accordingly rises from 0.49 to 0.64–0.67; at this accuracy ECE mostly restates mean confidence, so we read it as a rise in confidence without a matching rise in accuracy. The RL-trained medical models (AlphaMed; HuatuoGPT-o1 after SFT) thus state much higher confidence where they are least accurate. The effect also appears without instruction tuning: AlphaMed's base-initialized checkpoint is the most overconfident Qwen-family model (mean confidence 0.71, ECE 0.70, accuracy 1.7%). The medical descendants do gain a little accuracy: AlphaMed beats its parent at every $k$ (+0.6 [+0.2, +0.9] points at $k$=1, +1.0 [+0.1, +2.1] at $k$=32), HuatuoGPT-o1 at $k \leq 8$, and m1 at $k \geq 2$ (+1.4 [+0.4, +2.5] at $k$=32). These gains are an order of magnitude smaller than the rise in confidence. m1, trained only with supervised reasoning traces (at its intended 8k-token budget), is the exception on calibration and diversity: it is the most diverse model (15.3 distinct answers per 32 samples, against 11.5 for the parent and 9.2 for HuatuoGPT-o1), and its confidence rises least (0.57).

General post-training reproduces the DDXPlus pattern on Qwen3. Qwen3-8B beats its base at pass@1 (+0.7 [+0.4, +1.1]) and trails it from $k$=8 on, by −2.2 [−3.1, −1.3] points at pass@32 (a relative loss of 37%), with half the distinct answers (6.4 vs. 13.4). For Qwen2.5 no pass@$k$ difference is significant. Unlike medical RL, general post-training *lowers* stated confidence here (0.61 to 0.46 for Qwen3, 0.61 to 0.51 for Qwen2.5), although base models also give invalid answers more often (8.5% of Qwen3-8B-Base samples). OLMo 3 does not share this: every stage of its pipeline states a mean confidence of 0.73–0.84 at about 1% accuracy (pass@32 2.5–4.5%), and its SFT checkpoint is the most overconfident model we evaluate (confidence 0.84, ECE 0.83). On the 500-case subset, our 8B GRPO runs (§5) are indistinguishable from their base (all pass@$k$ differences within ±0.6 points).

Accuracy depends strongly on when the source case report was published (Table 19). For every model, pass@32 is 5–10% on cases published up to 2022 ($n$=901), at most 1.7% on 2023–24 cases ($n$=348), and 2.6–4.6% on 2025+ cases ($n$=152). The pattern is not monotone, since 2025+ cases score above 2023–24 cases, so case mix also matters; the gap between older and recent cases is consistent with, but does not establish, memorization of older case reports.

| Model | Single-call correct | Vote@32 gain | Top-3: list − 32 samples | AUROC, single-call answer correct | | | Correct answers |
|---|---|---|---|---|---|---|---|
| | | | | Stated $p$ | Agree@5 | Agree@32 | |
| Qwen3-8B-Base | 0.074 | +4.6 | −7.0 | 0.567 | 0.768 | **0.830** | 104 |
| Qwen3-8B | 0.080 | +0.4 | +0.4 | 0.707 | 0.726 | **0.763** | 112 |
| Qwen2.5-7B | 0.062 | +3.8 | −5.4 | 0.598 | 0.788 | **0.845** | 87 |
| Qwen2.5-7B-Instruct | 0.056 | +1.5 | −3.6 | 0.717 | 0.661 | **0.745** | 78 |
| OLMo-3-1025-7B | 0.060 | +3.1 | −4.6 | 0.464 | 0.765 | **0.814** | 84 |
| SFT | 0.035 | +1.1 | −3.6 | 0.628 | 0.730 | **0.814** | 49 |
| SFT+DPO | 0.026 | +1.2 | −3.3 | 0.651 | 0.673 | **0.725** | 36 |
| SFT+DPO+RLVR (Instruct) | 0.035 | +0.9 | −2.7 | 0.652 | 0.644 | **0.726** | 49 |
| RL-Zero (from base) | 0.065 | +2.3 | −3.1 | 0.583 | 0.791 | **0.827** | 91 |
| AlphaMed-7B | 0.071 | +1.6 | −3.0 | 0.586 | 0.755 | **0.797** | 100 |
| HuatuoGPT-o1-7B | 0.093 | +2.6 | −5.7 | 0.653 | 0.708 | **0.756** | 130 |
| m1-7B | 0.084 | +2.9 | −7.9 | 0.650 | 0.747 | **0.800** | 117 |
| AlphaMed-7B (base-RL) | 0.054 | +2.5 | −4.2 | 0.672 | 0.733 | **0.789** | 75 |

Table 20: Aggregation and uncertainty on Phenopacket Store (1,401 real rare-disease cases, 32 samples, $T$=0.7). An answer counts as correct if it is the diagnosis or a closely related Mondo class. Columns as in Table 8; Correct answers: number of correct single-call answers, the positives for the AUROCs. Bold: best signal.

### 4.18 Stated and sampled readouts on real rare-disease cases

Phenopacket Store cases are real, but they have no reference differential and accuracy is near the floor, so we can test only the consequences of §4.7, and we count an answer as correct if it is the diagnosis or a closely related Mondo class (exact matches give only 8–31 correct single-call answers per model). Top answers are grouped by Mondo class for voting and agreement (Table 20). The DDXPlus shifts reappear, but the top-three comparison does not flip as it does on DDXPlus. Majority voting over 32 samples helps the base models by 3.1–4.6 points but the final post-trained models by only 0.4–1.5 (Qwen3-8B +0.4, n.s.). Post-training weakens agreement as an error signal (Qwen3-8B 0.830 → 0.763, Qwen2.5-7B 0.845 → 0.745, OLMo 3 7B 0.814 → 0.726) and strengthens the stated probability (0.567 → 0.707, 0.598 → 0.717, 0.464 → 0.652). Agreement beats the stated probability by 0.25–0.35 AUROC for the base models and by only 0.03–0.07 for the post-trained ones, significantly only for Qwen3-8B ($-0.056$ $[-0.120, -0.002]$). On these open-ended cases the three most frequent of 32 sampled answers still contain a correct diagnosis more often than one list's top three, but the gap shrinks from 4.6–7.0 points for the base models to 2.7–3.6 for Qwen2.5-7B-Instruct and OLMo-3-7B-Instruct and to a tie for Qwen3-8B ($+0.4$ $[-0.8, +1.8]$). The medical models behave like their post-trained parent, with agreement the better signal for all of them.

## 5 Training without collapsing the samples

*Related rewards.* Closely related GRPO rewards computed from group answer counts have appeared since these experiments: a witness advantage that trains models to sample the distributions they state [1], and a batch bonus that penalizes repeated answers [31]. Our group reward `gm` additionally needs a reference differential at training time, which real clinical data rarely provide; we do not compare against them, and we treat the results below as existence results.

The stated list survives post-training, but sampled diversity still matters: self-consistency and other sampling-based methods depend on it (§4.7), and so does reinforcement learning itself, since GRPO learns nothing from a group whose samples agree. Can reinforcement learning raise diagnostic accuracy without collapsing the sampled differential? We fine-tune Qwen3-1.7B with GRPO

on DDXPlus training patients (9,800 cases, 200 per condition; disjoint from the test split) and evaluate exactly as in §3.

**Setup.** We train LoRA adapters ($r$=32) with GRPO for 200 steps. Each step uses 16 prompts and 16 sampled completions per prompt, generated with vLLM on the same GPU. A first sweep with a common aggressive configuration (learning rate $2\times10^{-5}$, no KL penalty, 8 samples per prompt, per-group reward standardization) was unstable. In three of seven runs, the policy collapsed to an input-independent answer: one condition for 88–100% of all patients, with completions shrinking to about 25 tokens. The starting policy is nearly deterministic (token entropy ≈0.06), so most GRPO groups have zero reward variance, and noisy updates drive the collapse. We therefore use learning rate $5\times10^{-6}$, KL coefficient 0.04, no reward standardization (as in Puri et al. [27]), and 16 samples per prompt. A monitor tracks the number of distinct top-1 answers per case within each batch. All runs in this configuration were stable.

**Rewards.** Every recipe includes a format reward: the answer must parse into a list of distinct conditions with valid probabilities. `acc` is standard RLVR: 1 if the top-1 answer is the true pathology. `dist` adds two per-sample distributional terms to `acc`: a Brier score of the stated distribution against the truth, and the overlap of the stated distribution with the reference differential. Both are computed from a *single* completion. They cannot see that the policy gives the same answer in every sample, which is exactly the collapse of §4.2. `gm` adds to `acc` a *group-level* reward over the GRPO group of $n$ samples for a case. If sample $i$'s top-1 answer is $a$, and $k_a$ samples in the group give $a$,

$$r_i^{\mathrm{gm}} = \mathrm{clip}\big(n\, p_{\mathrm{ref}}(a) - (k_a - 1),\ 0,\ 1\big), \tag{1}$$

where $p_{\mathrm{ref}}$ is the reference differential. This is sample $i$'s marginal contribution to $n\sum_c \min(q_c, p_{\mathrm{ref}}(c))$, the overlap between the group's empirical answer distribution $q$ and the reference. An answer earns credit while its condition is under-represented relative to its reference mass, and none once it is over-represented. In contrast to multi-answer rewards that score a set inside one completion [27], `gm` acts on the distribution of *sampled* answers, the quantity that collapses. Because the group reward fires only when a group contains dissenting samples, we also test a boosted variant, `gm`$^+$: `gm` weight 3 and training temperature 1.2. As baselines we adapt the multi-answer rewards of Puri et al. [27] to our prompt. Multi-RLVR scores the share of listed conditions that belong to the reference differential, treated as a set of acceptable answers; Multi-RLCR adds their set-level Brier term on the per-answer probabilities. Both zero the reward for duplicate or invalid answers, as in the original, and use the same format reward, optimizer and settings as our recipes.

**Results.** Table 22 reports means over two seeds; intervals bootstrap cases and do not include seed-to-seed variation. All our recipes raise top-1 accuracy by 8–10 points; the multi-answer baselines barely move it (below). The per-sample recipes (`acc`, `dist`) slightly *reduce* diversity relative to the starting model, the sharpening documented in §4.1. The group reward removes this. `gm` matches the starting model's diversity, and `gm`$^+$ exceeds it while also being the most accurate recipe. Against `acc`, with paired bootstraps over cases averaged over seeds, `gm`$^+$ improves:

- pass@1 by +1.6 [+1.0, +2.3] and pass@32 by +1.7 [+1.0, +2.6] points;
- distinct answers by +0.10 [+0.07, +0.13] and entropy by +0.034 [+0.022, +0.047];
- coverage@32 by +1.5 [+1.0, +2.1] points, and TV by −0.014 [−0.019, −0.010].

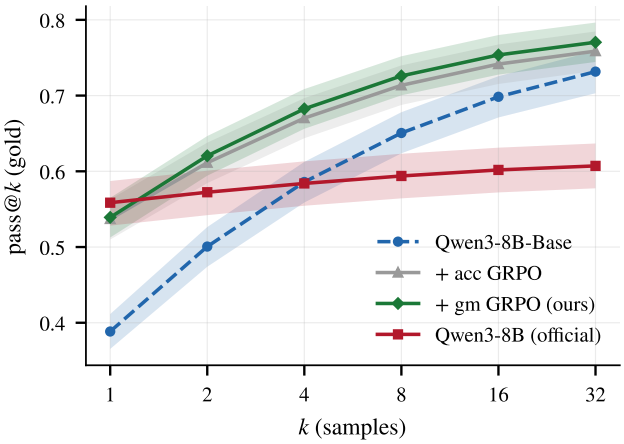



Figure 5: pass@$k$ on DDXPlus ($T$=0.7, 980 cases) from the same pretrained weights: Qwen3-8B-Base, our GRPO runs (pooled over two seeds), and the official post-trained Qwen3-8B, with 95% case-bootstrap bands. Our RL lifts the whole curve; official post-training trades the right end for the left.

Plain `gm` improves on `acc` in the same direction with smaller margins (e.g. pass@1 +1.0 [+0.6, +1.6], entropy +0.013 [+0.005, +0.021]). Diversity increases monotonically with group-reward pressure, with no accuracy cost. The set-membership baselines of Puri et al. [27] train stably (their reward rises from 0.27–0.31 to 0.32–0.35) but barely change sampled behavior. pass@1 rises by under 1 point over the untrained model, and diversity is unchanged. Against them, `gm`$^+$ is better on every metric, by +8.9 [+7.4, +10.5] points at pass@1, +0.07 [+0.03, +0.11] distinct answers and +3.0 [+2.3, +3.8] points of coverage@32 (vs. MULTI-RLVR; MULTI-RLCR is similar). What they do improve is the differential written *inside* one completion: the reference mass covered by a single listed differential rises to 0.317 (MULTI-RLVR), against 0.300 for `acc` and 0.309 for `gm`$^+$. The two objectives are complementary: a set reward shapes the list a model writes, while the group reward shapes which answer it commits to across samples, which is what collapses under post-training. The absolute effects are modest because the starting policy is already collapsed: there is little exploration for any reward to act on. None of the recipes changes rare-disease accuracy or confidence on Phenopacket Store, where a 1.7B model is at the floor; the same holds at 8B (§4.17).

**Starting from a diverse base model.** The 1.7B experiment starts from an already collapsed instruct model. We therefore repeat the comparison from Qwen3-8B-Base, the diverse base model of §4.1, with the same recipe and `acc` vs. `gm`. We compare against Qwen's own post-trained Qwen3-8B (Table 21; means of two seeds). Both RL recipes reach nearly the official model's top-1 accuracy (0.537 and 0.539 vs. 0.558; `gm`: −1.9 [−3.9, +0.1] points). They keep more than twice its diversity (2.7–2.9 vs. 1.27 distinct answers) and beat it by 15–16 points at pass@32 (`gm`: +16.3 [+14.0, +18.8]). In this setting, then, the collapse produced by general post-training is not required for its top-1 accuracy (our runs train on in-distribution DDXPlus data, so this is an existence result, not a like-for-like comparison of recipes). Both runs still narrow the base model somewhat, and the group reward narrows it less. At equal pass@1 (+0.2, n.s.), `gm` beats `acc` on distinct answers +0.14 [+0.11, +0.17], entropy +0.035 [+0.029, +0.041], coverage@32 +1.1 [+0.8, +1.5] points, and pass@32 +1.2 [+0.5, +1.9]. Each seed reproduces the ordering. The severe tier tells the same story (Table 3): official post-training removes 9.8 points of can't-miss consideration, `acc` 2.8, and `gm` 1.5 (not significant). Figure 5 shows the pass@$k$ curves.

| Model | pass@1 | pass@32 | Distinct | Entropy | Cov@32 | TV↓ |
|---|---|---|---|---|---|---|
| Qwen3-8B-Base (start) | 0.389 | 0.732 | **3.77** | **0.78** | **0.323** | 0.749 |
| + `acc` GRPO (ours) | 0.537 | 0.759 | 2.72 | 0.52 | 0.297 | 0.750 |
| + `gm` GRPO (ours) | 0.539 | **0.770** | 2.86 | 0.56 | 0.308 | **0.742** |
| Qwen3-8B (official post-training) | **0.558** | 0.607 | 1.27 | 0.11 | 0.217 | 0.793 |

Table 21: GRPO from Qwen3-8B-Base vs. the official post-trained Qwen3-8B on DDXPlus (980 cases, 32 samples, $T$=0.7); GRPO rows are means of two seeds. Our RL nearly matches the official model's top-1 accuracy while keeping more than twice its diversity; the group reward preserves more than `acc`. The base row comes from this experiment's own sampling run, so it differs slightly from Table 2.

| Recipe | pass@1 | pass@32 | Distinct | Entropy | Cov@32 | TV↓ |
|---|---|---|---|---|---|---|
| Qwen3-1.7B (untrained) | 0.378 | 0.435 | 1.41 | 0.151 | 0.162 | 0.850 |
| `acc` | 0.458 | 0.499 | 1.36 | 0.142 | 0.181 | 0.829 |
| `dist` | 0.458 | 0.502 | 1.34 | 0.131 | 0.181 | 0.829 |
| MULTI-RLVR [27] | 0.385 | 0.443 | 1.39 | 0.153 | 0.166 | 0.845 |
| MULTI-RLCR [27] | 0.383 | 0.441 | 1.39 | 0.150 | 0.165 | 0.847 |
| `gm` | 0.469 | 0.510 | 1.40 | 0.155 | 0.186 | 0.825 |
| `gm`$^+$ | **0.474** | **0.516** | **1.46** | **0.176** | **0.196** | **0.815** |

Table 22: GRPO on Qwen3-1.7B, evaluated on DDXPlus test (980 cases, 32 samples, $T$=0.7); means of two seeds. `gm`$^+$: group-reward weight 3, training temperature 1.2. Per-sample rewards slightly narrow the model; set-membership rewards barely move it; the group reward preserves, then widens, its differential while improving accuracy.

## 6 Condition-clustered intervals and multiple comparisons

DDXPlus has 20 test cases for each of 49 true pathologies, and effects can be specific to a condition, so cases are not independent. Table 23 gives the headline base-vs-post differences with a bootstrap that resamples the 49 pathologies with all their cases (10,000 resamples). Two-sided p-values use the add-one correction, $2(1+\min(n_{\leq 0}, n_{\geq 0}))/(B+1)$, so the smallest attainable value, 2/10,001, lies below the strictest Holm threshold, 0.05/112. Holm correction runs over the 112 tests (8 pairs × 14 metrics fixed in advance: pass@1, pass@32, distinct answers, coverage@32, sampled TV, list TV, mass named, severe named, list and sample TV without the true pathology, DDF1, unwarranted severe conditions named, and the two AUROCs; the table shows seven). The same procedure is applied separately to the few-shot pairs (70 tests), the Mistral and thinking-mode pairs (42) and the stage and medical comparisons (154). Not surviving correction among the headline tests, marked †: the 32B pass@32 losses, the sampled-TV increases for Qwen2.5-7B and OLMo 3.1 32B, OLMo 3 7B's list TV at both temperatures and its list TV without the true pathology at $T$=1, Qwen3-8B's list TV without the true pathology at $T$=0.7, OLMo 3 7B's DDF1 at $T$=0.7, and the drop in agreement AUROC for both OLMo models. OLMo 3 7B's list TV without the true pathology at $T$=0.7 survives as a significant *increase*.

## 7 Prompts, vignettes and grading

**DDXPlus.** Each patient's coded evidence is rendered by a fixed template into a vignette of the patient's answers and relevant history; we fix known mistranslations in the English evidence descrip-

| Pair (post − base) | pass@32 (pts) | Distinct | TV samples | TV one list | TV list w/o gold | DDF1 | AUROC agree@32 |
|---|---|---|---|---|---|---|---|
| Qwen3-8B, $T$=0.7 | −12.4 | −2.50 | +0.043 | −0.043 | −0.018† | +0.042 | −0.160 |
| | [−19.5, −5.9] | [−2.88, −2.13] | [+0.021, +0.063] | [−0.059, −0.028] | [−0.032, −0.005] | [+0.029, +0.056] | [−0.216, −0.097] |
| Qwen3-8B, $T$=1 | −14.6 | −5.06 | +0.067 | −0.074 | −0.039 | +0.066 | −0.158 |
| | [−21.6, −8.4] | [−5.49, −4.66] | [+0.042, +0.091] | [−0.090, −0.059] | [−0.052, −0.027] | [+0.053, +0.080] | [−0.211, −0.103] |
| Qwen2.5-7B, | −9.7 | −3.09 | +0.028† | −0.062 | −0.041 | +0.062 | −0.106 |
| $T$=0.7 | [−16.2, −4.1] | [−3.61, −2.58] | [+0.007, +0.049] | [−0.076, −0.049] | [−0.054, −0.028] | [+0.049, +0.076] | [−0.162, −0.041] |
| Qwen2.5-7B, $T$=1 | −10.3 | −5.59 | +0.042† | −0.088 | −0.056 | +0.083 | −0.104 |
| | [−17.4, −4.3] | [−6.04, −5.15] | [+0.015, +0.068] | [−0.103, −0.074] | [−0.069, −0.044] | [+0.069, +0.097] | [−0.160, −0.046] |
| Qwen2.5-32B, | −4.9† | −2.10 | +0.033 | −0.053 | −0.043 | +0.051 | −0.079 |
| $T$=0.7 | [−10.2, +0.6] | [−2.50, −1.74] | [+0.018, +0.049] | [−0.064, −0.042] | [−0.053, −0.033] | [+0.041, +0.061] | [−0.119, −0.039] |
| OLMo 3 7B, | −28.3 | −2.56 | +0.090 | +0.029† | +0.034 | −0.006† | −0.076† |
| $T$=0.7 | [−35.7, −21.2] | [−2.84, −2.27] | [+0.072, +0.109] | [+0.004, +0.052] | [+0.017, +0.051] | [−0.022, +0.009] | [−0.156, +0.000] |
| OLMo 3 7B, $T$=1 | −15.2 | −3.95 | +0.079 | −0.009† | +0.006† | +0.024 | −0.076† |
| | [−20.9, −9.6] | [−4.22, −3.68] | [+0.060, +0.098] | [−0.030, +0.011] | [−0.006, +0.019] | [+0.010, +0.036] | [−0.150, −0.009] |
| OLMo 3.1 32B, | −4.0† | −2.18 | +0.020† | −0.045 | −0.023 | +0.040 | −0.056† |
| $T$=0.7 | [−6.8, −1.2] | [−2.46, −1.89] | [+0.002, +0.037] | [−0.059, −0.031] | [−0.034, −0.012] | [+0.027, +0.053] | [−0.095, −0.018] |

Table 23: Base → post-trained (final Instruct) differences on DDXPlus with 95% intervals from a bootstrap over the 49 true-pathology clusters. †: not significant after Holm correction across 112 tests (10,000 resamples, add-one p-values). pass@32 here is recomputed from the samples with the analysis scripts' name matching and can differ from Table 2 by up to 0.3 points.

tions. An example (test case with true pathology chronic rhinosinusitis; the reference differential gives viral pharyngitis 0.17, chronic rhinosinusitis 0.16, cluster headache 0.15, acute rhinosinusitis 0.14, bronchitis 0.14, URTI 0.12):

> `A 13-year-old female presents for evaluation. Symptoms (patient answers): Do you have pain somewhere, related to your reason for consulting? Yes. Characterize your pain: burning. Do you feel pain somewhere? mouth, forehead, cheek(R), cheek(L), nose. How intense is the pain? (scale 0-10) 3. […] Have you lost your sense of smell? Yes. Do you have nasal congestion or a clear runny nose? Yes. Do you have a cough? Yes. Relevant history and risk factors: Have you had a cold in the last 2 weeks? Yes. […] Do you have polyps in your nose? Yes. […]`

System prompt: `You are an experienced emergency and primary care physician.` The user turn is the vignette followed by: `Based on the patient's answers above, give your differential diagnosis. Choose ONLY from this list of possible conditions: [the 49 conditions, one per line]. List the 5 most likely conditions from the list, most likely first, with your probability that each is the correct diagnosis. Use the condition names exactly as written in the list. End your answer with a JSON array in exactly this format: [{"diagnosis": "<name>", "probability": <number between 0 and 1>}, ...].` The full-distribution prompt (§4.8) instead asks for a probability on every condition, summing to one, most likely first. The plain and CoT variants are described in §4.3. Base models receive the same chat-formatted prompt; OLMo base models use their SFT checkpoint's chat template.

**Phenopacket Store.** The vignette lists the case's present and (up to ten) excluded HPO phenotypes with sex and age. System prompt: `You are an expert clinical geneticist and diagnostician.` Instructions: `List the 5 most likely specific diagnoses (use the specific disease or subtype name, e.g. 'Spastic paraplegia 7' rather than 'hereditary spastic paraplegia', when you can), most likely first, with your probability that each is the correct diagnosis`, followed by the same JSON format.

**Grading open-ended names.** Each predicted name is mapped to Mondo by exact lookup against Mondo labels and synonyms (also trying any parenthesized abbreviation and the name without it), then by fuzzy matching (token-sort ratio $\geq 92$). It is *exact* if it maps to the Mondo class of the case's OMIM diagnosis, and *related* if it maps to a specific ancestor group of it (at most two is-a steps up, with at most 150 descendants) or to one of its subtypes; siblings sharing only a broad parent are wrong. The grader passes hand-written checks (e.g. "Hereditary spastic paraplegia type 7 (SPG7)" is exact for SPG7, "Hereditary spastic paraplegia" is related, and Turner syndrome is wrong for Jacobsen syndrome), but we did not validate it against clinician grading, so errors at the scale of the 1–2 point effects we report on this dataset are possible.

## 8 Limitations

DDXPlus is synthetic: its reference differential is the posterior of a rule-based engine, and the simulated pathology is not always ranked first in it (it is for "more than 70%" of patients [8, §3.5]), so we treat TV and coverage as distance to a *reference* posterior rather than to clinical truth. Base models more often give invalid answers. We count these as errors, which makes the large-$k$ losses conservative but inflates post-trained gains at small $k$; conditioning on valid samples removes a quarter to a third of the $T$=1 pass@1 gain (§4.3). A re-run of the five final pairs with three worked examples in every prompt nearly removes invalid answers and confirms the results except OLMo 3.1 32B's stated-list gain (§4.10); it uses one temperature and one set of example patients, with lists from the reference or from Qwen2.5-32B-Instruct. Sampled hotter, Qwen2.5-7B-Instruct recovers much of its base model's pass@32 but Qwen3-8B does not (§4.11); the temperature control covers two 7–8B models. Our base/post-trained comparisons cover two open families at 7–32B, plus Mistral-7B with few-shot prompts only; prompt robustness of the sampled readout is tested on one pair, and the collapse at large $k$ largely disappears when the model is asked to reason first (§4.3) or uses its thinking mode (§4.13). The OLMo base models give invalid answers on 27–47% of samples, so their format-controlled comparisons are weaker than Qwen's (below 0.5% with few-shot prompts). We evaluate one RL-Zero checkpoint, at 7B, trained with an LM-judge reward on chat prompts rather than a diagnosis reward, and the 32B models only at $T$=0.7. Phenopacket cases are drawn from published literature and may be in pretraining data, which we partly address with era splits. The stated readout renormalizes each list's probabilities over the conditions it names, and the single-call answer is a model's first valid sample. A full-distribution prompt gives the same picture (§4.8), but other elicitations may differ. Which readout is the better error signal is model-dependent; the audit of §4.15 makes it checkable with 100 labeled cases but does not remove the dependence, and we validate it only within DDXPlus. The test set is class-balanced (20 cases per condition), so accuracies do not reflect real prevalence, and the prompt lists all 49 conditions, so DDXPlus is a closed-set task. We do not evaluate clinical utility with clinicians.

## References


[1] Kasra Arabi, Virginia Smith, and Chhavi Yadav. Sample what you say: Aligning language models to sample the distributions they state, 2026.

[2] Junying Chen, Zhenyang Cai, Ke Ji, Xidong Wang, Wanlong Liu, Rongsheng Wang, and Benyou Wang. Towards medical complex reasoning with LLMs through medical verifiable problems. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 14552–14573, Vienna, Austria, 2025. Association for Computational Linguistics. doi: 10.18653/

v1/2025.findings-acl.751. arXiv version titled "HuatuoGPT-o1, Towards Medical Complex Reasoning with LLMs".

[3] Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde de Oliveira Pinto, Jared Kaplan, Harri Edwards, Yuri Burda, Nicholas Joseph, Greg Brockman, Alex Ray, Raul Puri, Gretchen Krueger, Michael Petrov, Heidy Khlaaf, Girish Sastry, Pamela Mishkin, Brooke Chan, Scott Gray, Nick Ryder, Mikhail Pavlov, Alethea Power, Lukasz Kaiser, Mohammad Bavarian, Clemens Winter, Philippe Tillet, Felipe Petroski Such, Dave Cummings, Matthias Plappert, Fotios Chantzis, Elizabeth Barnes, Ariel Herbert-Voss, William Hebgen Guss, Alex Nichol, Alex Paino, Nikolas Tezak, Jie Tang, Igor Babuschkin, Suchir Balaji, Shantanu Jain, William Saunders, Christopher Hesse, Andrew N. Carr, Jan Leike, Josh Achiam, Vedant Misra, Evan Morikawa, Alec Radford, Matthew Knight, Miles Brundage, Mira Murati, Katie Mayer, Peter Welinder, Bob McGrew, Dario Amodei, Sam McCandlish, Ilya Sutskever, and Wojciech Zaremba. Evaluating large language models trained on code. 2021.

[4] Zhoujun Cheng, Shibo Hao, Tianyang Liu, Fan Zhou, Yutao Xie, Feng Yao, Yuexin Bian, Yonghao Zhuang, Nilabjo Dey, Yuheng Zha, Yi Gu, Kun Zhou, Yuqi Wang, Yuan Li, Richard Fan, Jianshu She, Chengqian Gao, Abulhair Saparov, Haonan Li, Taylor W. Killian, Mikhail Yurochkin, Zhengzhong Liu, Eric P. Xing, and Zhiting Hu. Revisiting reinforcement learning for LLM reasoning from a cross-domain perspective. In *Advances in Neural Information Processing Systems (NeurIPS 2025), Datasets and Benchmarks Track*, 2025.

[5] Mehul Damani, Isha Puri, Stewart Slocum, Idan Shenfeld, Leshem Choshen, Yoon Kim, and Jacob Andreas. Beyond binary rewards: Training LMs to reason about their uncertainty. In *International Conference on Learning Representations (ICLR)*, 2026.

[6] Daniel Danis, Michael J. Bamshad, Yasemin Bridges, Andrés Caballero-Oteyza, Pilar Cacheiro, Leigh C. Carmody, Leonardo Chimirri, Jessica X. Chong, Ben Coleman, Raymond Dalgleish, Peter J. Freeman, Adam S. L. Graefe, Tudor Groza, Peter Hansen, Julius O. B. Jacobsen, Adam Klocperk, Maaike Kusters, Markus S. Ladewig, Allison J. Marcello, Teresa Mattina, Christopher J. Mungall, Monica C. Munoz-Torres, Justin T. Reese, Filip Rehburg, Bárbara C. S. Reis, Catharina Schuetz, Damian Smedley, Timmy Strauss, Jagadish Chandrabose Sundaramurthi, Sylvia Thun, Kyran Wissink, John F. Wagstaff, David Zocche, Melissa A. Haendel, and Peter N. Robinson. A corpus of GA4GH phenopackets: Case-level phenotyping for genomic diagnostics and discovery. *Human Genetics and Genomics Advances*, 6(1):100371, 2025. doi: 10.1016/j.xhgg.2024.100371.

[7] Arsene Fansi Tchango, Rishab Goel, Julien Martel, Zhi Wen, Gaetan Marceau Caron, and Joumana Ghosn. Towards trustworthy automatic diagnosis systems by emulating doctors' reasoning with deep reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS 2022)*, volume 35, 2022.

[8] Arsene Fansi Tchango, Rishab Goel, Zhi Wen, Julien Martel, and Joumana Ghosn. DDXPlus: A new dataset for automatic medical diagnosis. In *Advances in Neural Information Processing Systems (NeurIPS 2022), Datasets and Benchmarks Track*, 2022.

[9] Sebastian Farquhar, Jannik Kossen, Lorenz Kuhn, and Yarin Gal. Detecting hallucinations in large language models using semantic entropy. *Nature*, 630(8017):625–630, 2024.

[10] Xiaoke Huang, Juncheng Wu, Hui Liu, Xianfeng Tang, and Yuyin Zhou. m1: Unleash the potential of test-time scaling for medical reasoning with large language models. In *Proceedings*

*of the Fifth Machine Learning for Health Symposium (ML4H 2025)*, volume 297 of *Proceedings of Machine Learning Research*, pages 369–383. PMLR, 2026.

[11] Chaemin Jang, Dongman Lee, and Jihee Kim. Instruction-tuned language models cannot sample from distributions they can describe. In *Findings of the Association for Computational Linguistics: EMNLP 2026*, 2026.

[12] Ahmadreza Jeddi, Kimia Shaban, Negin Baghbanzadeh, Natasha Sharan, Abhishek Moturu, Elham Dolatabadi, and Babak Taati. When does RL help medical VLMs? disentangling vision, SFT, and RL gains. 2026.

[13] Liwei Jiang, Yuanjun Chai, Margaret Li, Mickel Liu, Raymond Fok, Nouha Dziri, Yulia Tsvetkov, Maarten Sap, Alon Albalak, and Yejin Choi. Artificial hivemind: The open-ended homogeneity of language models (and beyond). In *Advances in Neural Information Processing Systems (NeurIPS 2025), Datasets and Benchmarks Track*, 2025.

[14] Zahir Kanjee, Byron Crowe, and Adam Rodman. Accuracy of a generative artificial intelligence model in a complex diagnostic challenge. *JAMA*, 330(1):78, 2023. doi: 10.1001/jama.2023.8288.

[15] Robert Kirk, Ishita Mediratta, Christoforos Nalmpantis, Jelena Luketina, Eric Hambro, Edward Grefenstette, and Roberta Raileanu. Understanding the effects of RLHF on LLM generalisation and diversity. In *International Conference on Learning Representations (ICLR)*, 2024.

[16] Thom Lake, Eunsol Choi, and Greg Durrett. From distributional to overton pluralism: Investigating large language model alignment. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 6794–6814, 2025.

[17] Jixuan Leng, Chengsong Huang, Banghua Zhu, and Jiaxin Huang. Taming overconfidence in LLMs: Reward calibration in RLHF. In *International Conference on Learning Representations (ICLR)*, 2025.

[18] Seungseop Lim, Gibaeg Kim, Hyunkyung Lee, Wooseok Han, Jean Seo, Jaehyo Yoo, and Eunho Yang. H-DDx: A hierarchical evaluation framework for differential diagnosis. 2025. NeurIPS 2025 Workshop on Generative AI for Health (GenAI4Health).

[19] Che Liu, Haozhe Wang, Jiazhen Pan, Zhongwei Wan, Yong Dai, Fangzhen Lin, Wenjia Bai, Daniel Rueckert, and Rossella Arcucci. Beyond distillation: Pushing the limits of medical LLM reasoning with minimalist rule-based RL. 2025. Introduces the AlphaMed model; NeurIPS 2025 Workshop on Generative AI for Health (GenAI4Health).

[20] Mingyi Liu. The alignment tax: Response homogenization in aligned LLMs and its implications for uncertainty estimation, 2026.

[21] Qing Lyu, Kumar Shridhar, Chaitanya Malaviya, Li Zhang, Yanai Elazar, Niket Tandon, Marianna Apidianaki, Mrinmaya Sachan, and Chris Callison-Burch. Calibrating large language models with sample consistency, 2024.

[22] Daniel McDuff, Mike Schaekermann, Tao Tu, Anil Palepu, Amy Wang, Jake Garrison, Karan Singhal, Yash Sharma, Shekoofeh Azizi, Kavita Kulkarni, et al. Towards accurate

differential diagnosis with large language models. *Nature*, 642(8067):451–457, 2025. doi: 10.1038/s41586-025-08869-4.

[23] Zhiting Mei, Christina Zhang, Tenny Yin, Justin Lidard, Ola Shorinwa, and Anirudha Majumdar. Reasoning about uncertainty: Do reasoning models know when they don't know? In *Findings of the Association for Computational Linguistics: EACL 2026*, pages 3408–3458, Rabat, Morocco, 2026. Association for Computational Linguistics. doi: 10.18653/v1/2026.findings-eacl.178.

[24] Nicole Meister, Carlos Guestrin, and Tatsunori Hashimoto. Benchmarking distributional alignment of large language models. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 24–49, Albuquerque, New Mexico, 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.naacl-long.2.

[25] Sonia Krishna Murthy, Tomer Ullman, and Jennifer Hu. One fish, two fish, but not the whole sea: Alignment reduces language models' conceptual diversity. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 11241–11258, 2025. doi: 10.18653/v1/2025.naacl-long.561.

[26] Changdae Oh, Qi Zeng, Qi Qi, Andrey Zhmoginov, Deren Lei, Yun He, Hoang Phan, Hangoo Kang, Azalia Mirhoseini, and Sharon Li. Sharpening tax in post-training. 2026.

[27] Isha Puri, Mehul Damani, Idan Shenfeld, Marzyeh Ghassemi, Jacob Andreas, and Yoon Kim. Escaping the mode: Multi-answer reinforcement learning in LMs. In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, volume 306 of *Proceedings of Machine Learning Research*, pages 100127–100146. PMLR, 2026. arXiv version titled "Reaching Beyond the Mode: RL for Distributional Reasoning in Language Models".

[28] Zhiyao Ren, Yibing Zhan, Siyuan Liang, Guozheng Ma, Baosheng Yu, and Dacheng Tao. Towards reliable medical LLMs: Benchmarking and enhancing confidence estimation of large language models in medical consultation, 2026.

[29] Daniel Philip Rose, Chia-Chien Hung, Marco Lepri, Israa Alqassem, Kiril Gashteovski, and Carolin Lawrence. MEDDxAgent: A unified modular agent framework for explainable automatic differential diagnosis. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 13803–13826, 2025. doi: 10.18653/v1/2025.acl-long.677.

[30] Thomas Savage, John Wang, Robert Gallo, Abdessalem Boukil, Vishwesh Patel, Seyed Amir Ahmad Safavi-Naini, Ali Soroush, and Jonathan H. Chen. Large language model uncertainty proxies: discrimination and calibration for medical diagnosis and treatment. *Journal of the American Medical Informatics Association*, 32(1):139–149, 2025. doi: 10.1093/jamia/ocae254.

[31] Yuda Song, Julia Kempe, and Remi Munos. Outcome-based exploration for LLM reasoning, 2025.

[32] Jacob Mitchell Springer, Madhu Advani, Lukas Aichberger, Arwen Bradley, Eran Malach, Omid Saremi, Sinead Williamson, Preetum Nakkiran, Etai Littwin, and Aditi Raghunathan.

Annotations mitigate post-training mode collapse. In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, 2026.

[33] Team Olmo, Allyson Ettinger, Amanda Bertsch, Bailey Kuehl, David Graham, David Heineman, Dirk Groeneveld, Faeze Brahman, Finbarr Timbers, Hamish Ivison, et al. Olmo 3, 2025.

[34] Katherine Tian, Eric Mitchell, Allan Zhou, Archit Sharma, Rafael Rafailov, Huaxiu Yao, Chelsea Finn, and Christopher D. Manning. Just ask for calibration: Strategies for eliciting calibrated confidence scores from language models fine-tuned with human feedback. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023.

[35] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations (ICLR)*, 2023.

[36] Peter West and Christopher Potts. Base models beat aligned models at randomness and creativity. In *Conference on Language Modeling (COLM)*, 2025.

[37] Fang Wu, Weihao Xuan, Danny Xing, Ximing Lu, Mingjie Liu, Yi Dong, Zaid Harchaoui, and Yejin Choi. The invisible leash: Why RLVR may or may not escape its origin. 2025.

[38] Miao Xiong, Zhiyuan Hu, Xinyang Lu, Yifei Li, Jie Fu, Junxian He, and Bryan Hooi. Can LLMs express their uncertainty? an empirical evaluation of confidence elicitation in LLMs. In *International Conference on Learning Representations (ICLR)*, 2024.

[39] Yang Yue, Zhiqi Chen, Rui Lu, Andrew Zhao, Zhaokai Wang, Yang Yue, Shiji Song, and Gao Huang. Does reinforcement learning really incentivize reasoning capacity in LLMs beyond the base model? In *Advances in Neural Information Processing Systems (NeurIPS 2025)*, 2025.

[40] Jiayi Zhang, Simon Yu, Derek Chong, Anthony Sicilia, Michael R. Tomz, Christopher D. Manning, and Weiyan Shi. Verbalized sampling: How to mitigate mode collapse and unlock LLM diversity. In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, volume 306 of *Proceedings of Machine Learning Research*, pages 160073–160152. PMLR, 2026.